

C2 Assignment : Integral Calculus

SINGLE CORRECT TYPE QUESTIONS

I04/1/001. The area of the loop formed by $y^2 = x(1 - x^3)$ is :

- (A) $\int_0^1 \sqrt{x - x^4} dx$ (B) $2 \int_0^1 \sqrt{x - x^4} dx$ (C) $\int_{-1}^1 \sqrt{x - x^4} dx$ (D) $4 \int_{-1}^{\frac{1}{2}} \sqrt{x - x^4} dx$

Ans. B

Sol. Area = $2 \int_0^1 \sqrt{x - x^4} dx$

I04/1/002. Given $y = f(x)$ is a solution of differential equation $xydy + xdx = ydy - ydx$, also $f(0) = 0$, $f(1) = \tan \frac{3\pi}{8}$, then

the area bounded by $y = f(x)$ above x-axis and $x = 0$ to $x = \tan \frac{\pi}{8}$ is equal to :

- (A) $\frac{1}{2}(\sqrt{2} - 1)$ (B) $\frac{1}{2}(\sqrt{2} + 1)^3$ (C) $\frac{1}{2}(\sqrt{2} - 1)^3$ (D) $\frac{1}{2}(\sqrt{2} + 1)^2$

Ans. A

Sol. $xydy + ydx = ydy - xdx$

$$\Rightarrow d(xy) = ydy - xdx$$

$$\Rightarrow 2xy = y^2 - x^2 + c$$

...(1)

Put $x = 0, y = 0$, we get $c = 0$

$$\therefore (1) \Rightarrow y^2 - 2xy - x^2 = 0$$

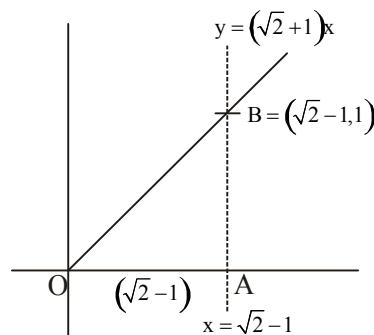
$$\Rightarrow y = 2x \pm \sqrt{4x^2 + 4x^2}$$

$$y = x + \sqrt{2}|x|$$

$$\text{Or } y = x - \sqrt{2}|x|$$

$$\text{But } y(1) = \tan \frac{3\pi}{8} \Rightarrow y = x + \sqrt{2}|x|$$

$$\therefore \text{ Required area} = \frac{1}{2}(\sqrt{2} - 1) = \frac{1}{2}(\sqrt{2} - 1)$$



I04/1/003. Consider the curves $x^2 + y^2 - 4x - 6y - 12 = 0$, $9x^2 + 4y^2 - 900 = 0$ and $y^2 - 6y - 6x + 51 = 0$. The maximum number of disjoint regions into which these curves divide the XY-plane (excluding the curves

themselves), is :

- (A) 4 (B) 5 (C) 6 (D) 7

Ans. B

Sol. The equation of the first curve is

$$(x - 2)^2 + (y - 3)^2 = 25.$$

This is a circle with centre (2, 3) and radius 5.

The equation of the second curve is

$$9x^2 + 4y^2 = 900.$$

This is an ellipse. The first curve is entirely inside this ellipse. Indeed, if

$$(x - 2)^2 + (y - 3)^2 = 25,$$

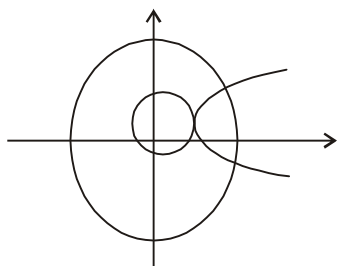
then $|x - 2| \leq 5$, $|y - 3| \leq 5$, hence $|x| \leq 7$, $|y| \leq 8$ and

$$9x^2 + 4y^2 \leq 697 < 900.$$

The first two curves divide the plane into 3 regions. The equation of the third curve is

$$6x = (y - 3)^2 + 42.$$

This is parabola with vertex (7, 3) and which is rightward opened. Observe that the first curve and the third curve touch each other at the point (3, 7) which is the rightmost point of the first curve. Hence, the third curve does not divide the region of the circle inside the third curve, whhiel it divides two other regions into two parts. Totally, we have 5 regions.



I04/1/004. Consider the functions $f, g : [0, 1] \rightarrow [0, 1]$ given by $f(x) = \frac{1}{2}x(x+1)$ and $g(x) = \frac{1}{2}x^2(x+1)$. Then the area enclosed between the graphs of f^{-1} and g^{-1} is :

- (A) $\frac{1}{4}$ (B) $\frac{1}{6}$ (C) $\frac{1}{8}$ (D) $\frac{1}{24}$

Ans. C

Sol. When you take inverse of a function, it just means you are reflectig it along the line $y = x$. Reflection does not change the area under the curve. so area between two curves is same as the area between the inverses of two curves. So

$$\text{Area between two curves} = \int_0^1 g^{-1}(x) - f^{-1}(x) dx = \int_0^1 g(x) - f(x) dx = \left(\frac{1}{2}\right) \times \left(\frac{1}{2} - \frac{1}{4}\right) = \frac{1}{8}.$$

So, C is the correct option

I02/1/005. The value of the constant $a > 0$ such that $\int_0^a \left[\tan^{-1} \sqrt{x} \right] dx = \int_0^a \left[\cot^{-1} \sqrt{x} \right] dx$, where $[.]$ denotes the greatest integer function is :

- (A) $\frac{2(3 + \cos 4)}{(1 - \cos 4)}$ (B) $\frac{2(3 - \cos 4)}{(1 + \cos 4)}$ (C) $\frac{2(3 - \cos 4)}{(1 - \cos 4)}$ (D) $\frac{2(3 + \cos 4)}{(1 + \cos 4)}$

Ans. A

Sol. The given integrals can be equal only if $a > \tan^2 1$

$$\therefore \int_0^a \left[\tan^{-1} \sqrt{x} \right] dx = \int_0^{\tan^2 1} \left[\tan^{-1} \sqrt{x} \right] dx + \int_{\tan^2 1}^a \left[\tan^{-1} \sqrt{x} \right] dx = 0 + 1 \int_{\tan^2 1}^a dx = a - \tan^2 1$$

$$\int_0^a \left[\cot^{-1} \sqrt{x} \right] dx = \int_{\tan^2 1}^a \left[\cot^{-1} \sqrt{x} \right] dx + 0 = 1(\cot^2 1 - 0)$$

$$\therefore a - \tan^2 1 = \cot^2 1$$

$$\Rightarrow a = \tan^2 1 + \cot^2 1$$

$$= \frac{\sin^2 1}{\cos^2 1} + \frac{\cos^2 1}{\sin^2 1}$$

$$= \frac{1 - \cos 2}{1 + \cos 2} + \frac{1 + \cos 2}{1 - \cos 2} = \frac{(1 + \cos 2)^2 + (1 - \cos 2)^2}{\sin^2 2}$$

$$= \frac{2(1 + \cos^2 2)}{\sin^2 2}$$

$$= \frac{2(2 + 2\cos^2 2)}{2\sin^2 2}$$

$$= \frac{2(3 + \cos 4)}{1 - \cos 4}$$

I02/1/006. The integral part of $\sum_{n=2}^{999} \frac{1}{\sqrt{n}}$ equal :

- (A) 196 (B) 197 (C) 198 (D) 199

Ans. B

Sol. Compare the sum with the integral of $\frac{1}{\sqrt{x}}$:

$$\sum_{n=2}^{9999} \frac{1}{\sqrt{n}} = \sum_{n=2}^{9999} \int_{n-1}^n \frac{1}{\sqrt{x}} dx < \int_1^{9999} \frac{dx}{\sqrt{x}} = 2\sqrt{9999} - 2 < 2 \cdot 100 - 2 = 198$$

$$d \sum_{n=2}^{9999} \frac{1}{\sqrt{n}} = \sum_{n=2}^{9999} \int_n^{n+1} \frac{1}{\sqrt{x}} dx > \int_2^{10000} \frac{dx}{\sqrt{x}} = 2\sqrt{10000} - 2\sqrt{2} = 200 - 2\sqrt{2} > 197$$

Hence, the integer part of the sum is 197.

I02/1/007. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function such that

$$f(x+1) = \frac{1}{2} f(x) \text{ for all } x \in \mathbb{R},$$

and let $a_n = \int_0^n f(x) dx$ for all integers $n > 1$. Then :

(A) $\lim_{n \rightarrow \infty} a_n$ exists and equals $\int_0^1 f(x) dx$

(B) $\lim_{n \rightarrow \infty} a_n$ does not exist

(C) $\lim_{n \rightarrow \infty} a_n$ exists if and only if $\left| \int_0^1 f(x) dx \right| < 1$

(D) $\lim_{n \rightarrow \infty} a_n$ exists and equals $2 \int_0^1 f(x) dx$

Ans. D

Sol. We verify by induction that $f(x+k) = \frac{1}{2^k} f(x)$, $k > 0$. Indeed, for $k=0$ this is obvious. If the result is true for k , then

$$f(x+k+1) = \frac{1}{2} f(x+k) = \frac{1}{2^{k+1}} f(x)$$

Hence,

$$a_n = \int_0^n f(x) dx = \sum_{k=0}^{n-1} \int_k^{k+1} f(x) dx = \sum_{k=0}^{n-1} \int_0^1 f(x+k) dx$$

$$= \sum_{k=0}^{n-1} \frac{1}{2^k} \int_0^1 f(x) dx = \int_0^1 f(x) dx \sum_{k=0}^{n-1} \frac{1}{2^k} \rightarrow \int_0^1 f(x) dx \sum_{k=0}^{\infty} \frac{1}{2^k} = 2 \int_0^1 f(x) dx, n \rightarrow \infty$$

I02/1/008. Let $f(x) = e^{-|x|}$, $x \in \mathbb{R}$ and $g(\theta) = \int_{-1}^1 f\left(\frac{x}{\theta}\right) dx$, $\theta \neq 0$.

Then, $\lim_{\theta \rightarrow 0} \frac{g(\theta)}{\theta}$:

- (A) equals 0 (B) equals $+\infty$ (C) equals 2 (D) does not exist

Ans. D

Sol. Consider the transformation

$$g(\theta) = \int_{-1}^1 e^{-\frac{|x|}{|\theta|}} dx = 2 \int_0^1 e^{-\frac{x}{|\theta|}} dx = 2|\theta| \int_0^{\frac{1}{|\theta|}} e^{-x} dx$$

Hence,

$$\lim_{\theta \rightarrow 0^+} \frac{g(\theta)}{\theta} = \lim_{\theta \rightarrow 0^+} 2 \int_0^{\frac{1}{|\theta|}} e^{-x} dx = 2 \int_0^{\infty} e^{-x} dx = 2$$

$$\lim_{\theta \rightarrow 0^+} \frac{g(\theta)}{\theta} = - \lim_{\theta \rightarrow 0^+} 2 \int_0^{\frac{1}{|\theta|}} e^{-x} dx = -2 \int_0^{\infty} e^{-x} dx = -2$$

I02/1/009. Let

$$p(x) = x^3 - 3x^2 + 2x, x \in \mathbb{R},$$

$$f_0(x) = \begin{cases} \int_0^x p(t) dt, & x \geq 0 \\ -\int_x^0 p(t) dt, & x < 0 \end{cases}$$

$$f_1(x) = e^{f_0(x)}, f_2(x) = e^{f_1(x)}, \dots, f_n(x) = e^{f_{n-1}(x)}$$

How many roots does the equation $\frac{df_n(x)}{dx} = 0$ have in the interval $(-\infty, \infty)$?

- (A) 1 (B) 3 (C) $n + 3$ (D) $3n$

Ans. B

Sol. For all $n \leq 1$ we have $f'_n(x) = f_n(x) f'_{n-1}(x)$ and $f_n(x) > 0$. So, $f'_n(x) = 0$ if and only if $f'_{n-1}(x) = 0$. The

number of solutions of the equation $f'_n(x) = 0$ coincides with the number of solutions of the equation $f'_0(x) = 0$. $f'_0(x) = p(x)$, so we need to find the number of solutions of the equation $x^3 - 3x^2 + 2x = 0$. There are three solutions: $x = 0, 1, 2$.

I02/1/010. For $n \in \mathbb{N}$, let a_n be defined

$$a_n = \int_0^n \frac{1}{1+nx^2} dx$$

Then $\lim_{n \rightarrow \infty} a_n$:

- (A) equals 0 (B) equals $\frac{\pi}{4}$ (C) equals $\frac{\pi}{2}$ (D) does not exist

Ans. A

Sol.

$$\begin{aligned}
 a_n &= \int_0^n \frac{1}{1+nx^2} dx \\
 &= \frac{1}{\sqrt{n}} \int_0^n \frac{1}{1+(\sqrt{nx})^2} d(\sqrt{nx}) \\
 &= \frac{1}{\sqrt{n}} \left[\tan^{-1}(\sqrt{nx}) \right]_0^n \\
 a_n &= \frac{1}{\sqrt{n}} \tan^{-1}\left(n^{\frac{3}{2}}\right) \\
 &= \frac{1}{\sqrt{n}} \left(\frac{\pi}{2} - \cot^{-1}\left(n^{\frac{3}{2}}\right) \right) \\
 \Rightarrow a_n &= \frac{\pi}{2\sqrt{n}} - \frac{1}{\sqrt{n}} \tan^{-1}\left(\frac{1}{n^{\frac{3}{2}}}\right) \\
 &= \lim_{n \rightarrow \infty} \tan^{-1}\left(\frac{1}{n^{\frac{3}{2}}}\right) = \frac{1}{n^{\frac{3}{2}}}, \text{ as } \frac{1}{n^{\frac{3}{2}}} \rightarrow 0 \\
 \lim_{n \rightarrow \infty} a_n &= \frac{\pi}{2\sqrt{n}} - \frac{1}{n^2}
 \end{aligned}$$

$$\Rightarrow \lim_{n \rightarrow \infty} a_n = 0$$

I02/1/011. Let $X = \frac{1}{1001} + \frac{1}{1002} + \frac{1}{1003} + \dots + \frac{1}{3001}$. Then :

- (A) $X < 1$ (B) $X > \frac{3}{2}$ (C) $1 < X < \frac{3}{2}$ (D) None of the above holds

Ans. C

Sol. $H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} = \sum_{k=1}^n \frac{1}{k}$

Now, we need to find

$$H_{3001} - H_{1000}$$

For finding out approximate answer.

$\frac{1}{n}$ is monotonically decreasing function, so by using remain sum definite integral

$$\int_{1002}^{3002} \frac{1}{t} dt \leq \sum_{n=1001}^{3001} \frac{1}{n} \leq \int_{1001}^{3001} \frac{1}{t} dt$$

which simplifies to :

$$\ln \frac{3002}{1002} \leq \text{answer} \leq \ln \frac{3001}{1001}$$

This values can be approximated to :

$$1 < \text{answer} < 1.5 \Rightarrow 1 < X < \frac{3}{2}$$

I02/1/012. Consider the functions

$$f(x) = \begin{cases} \int_0^x \{5 + |1 - y|\} dy & \text{if } x > 2 \\ 5x + 2 & \text{if } x \leq 2 \end{cases}$$

Then :

- (A) f is not continuous at $x = 2$
 (B) f is continuous and differentiable everywhere
 (C) f is continuous everywhere but not differentiable at $x = 1$
 (D) f is continuous everywhere but not differentiable at $x = 2$

Ans. D

Sol.

$$f(x) = \begin{cases} \int_0^x 5 + |1-y| dy & \text{if } x > 2 \\ 5x + 2 & \text{if } x \leq 2 \end{cases}$$

$$f'(x) = \begin{cases} 5 + |1-x| & \text{if } x > 2 \\ 5 & \text{if } x \leq 2 \end{cases}$$

Using Newton-Leibnitz rule for differentiation for checking the differentiability:

$$f'(2^+) = \lim_{h \rightarrow 0} 5 + |1 - (2+h)| = 6$$

and

$$f'(2^-) = 5$$

Hence, the function is not differentiable ($\because f'(2^+) \neq f'(2^-)$)

I02/1/013. Let $\psi: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function with $\psi(y) = 0$ for all $y \notin [0, 1]$ and $\int_0^1 \psi(y) dy = 1$. Let

$f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function. Then the value of $\lim_{n \rightarrow \infty} n \int_0^{100} f(x) \psi(nx) dx$ is:

- (A) $f(0)$ (B) $f'(0)$ (C) $f''(0)$ (D) $f(100)$

Ans. A

Sol. Put $nx = t$, $ndx = dt$ and $x = \frac{t}{n}$ we get

$$\begin{aligned} \lim_{n \rightarrow \infty} n \int_0^{100} f(x) \psi(nx) dx &= \lim_{n \rightarrow \infty} \int_0^{100n} f\left(\frac{t}{n}\right) \psi(t) dt \\ &= \int_0^1 f(0) \psi(t) dt. \end{aligned}$$

The last step is really important $\frac{t}{n}$ approaches 0. $100n \gg 1$ and we don't have to go beyond 1 because $\psi(y) = 0$ elsewhere as given in question.

$$\text{Continuing on } = f(0) \int_0^1 \psi(t) dt = f(0) * 1 = f(0)$$

So A is correct.

I03/1/014. The solution of the differential equation $(1 + x^2y^2)ydx + (x^2y^2 - 1)xdy = 0$ is:

- (A) $xy = \log x - \log y + C$ (B) $xy = \log y - \log x + C$
(C) $x^2y^2 = 2(\log x - \log y) + C$ (D) $x^2y^2 = 2(\log y - \log x) + C$

Ans. D

Sol. This differential equation can be solved by using the method of inspection. "Rearranging the terms of the D.E., we get $xdy - ydx = (x^2y^2)(ydx + xdy)$

$$\Rightarrow \frac{xdy - ydx}{xy} = xy(xdy + ydx)$$

$$\Rightarrow d\left(\log\left(\frac{y}{x}\right)\right) = xyd(xy)$$

$$\Rightarrow \log\left(\frac{y}{x}\right) = \frac{x^2y^2}{2} + c$$

Rearranging the above equation, we get $x^2y^2 = 2(\log y - \log x) + C$

Option D is correct.

ONE OR MORE THAN ONE CORRECT TYPE QUESTIONS

I04/2/015. A curve $y = f(x)$, ($x \geq 0$) passes through the origin such that $f(1) = 2$. Through any point (x, y) on the curve, lines are drawn parallel to the co-ordinate axes. If the curve (function is concave downward) divides the area formed by these lines and coordinate axes in the ratio 1 : 2 then which of the following is/are correct :

(A) $f(x) > x$ for all $x \in (0, 4)$

(B) $f(x) < x$ for all $x \in (0, 4)$

(C) $f(x) > x$ for all $x \in (4, \infty)$

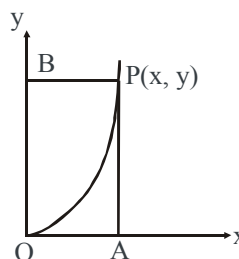
(D) $f(x) < x$ for all $x \in (4, \infty)$

Ans. CB

Sol. Area of OBPO : area of OPAO = m : n

$$\Rightarrow \frac{xy - \int_0^x ydx}{\int_0^x ydx} = \frac{m}{n}$$

$$\Rightarrow nxy = (m + n) \int_0^x ydx$$



Differentiating w.r.t x, we get

$$n\left(x \frac{dy}{dx} + y\right) = (m + n)y$$

$$\Rightarrow nx \frac{dy}{dx} = my \Rightarrow \frac{m}{n} \frac{dx}{x} = \frac{dy}{y}$$

$$\Rightarrow y = cx^{m/n} \text{ (put } m=1 \text{ and } n=2)$$

$$y = 2\sqrt{x}$$

I04/2/016. Let two curves $y = f(x)$ with $f(0) = 1$ and $g(x) = \int_{-\infty}^x f(t) dt$ with $g(0) = 4$ are such that, their tangents at the points with equal abscissa intersect on x-axis, then :

(A) $\int_{-1}^1 \frac{x^2 + 4}{4 + g(x)} dx = \frac{13}{3}$

(B) area bounded by $y = f(x)$, $y = g(x)$, $x = 1$ and y axis is $12(e^{1/4} - 1)$ sq. unit

(C) $f(x) - g(x)$ is decreasing in $[-1, 1]$

(D) period of $f\left(\left\{\frac{x}{2}\right\}\right)$ is 8 ($\{.\}$ denotes fractional part of x)

Ans. BC

Sol. $y - f(x) = f'(x)(y - x)$ and $y - g(x) = g'(x)(y - x)$

$$\Rightarrow \frac{d'(x)}{f(x)} = \frac{f'(x)}{g(x)} \Rightarrow f(x) = \frac{1}{4}g(x) \Rightarrow f(x) = e^{x/4}, g(x) = 4e^{x/4}$$

$$I = \int_{-1}^1 \frac{x^2 + 4}{4 + 4e^{x/4}} dx = \frac{13}{12}$$

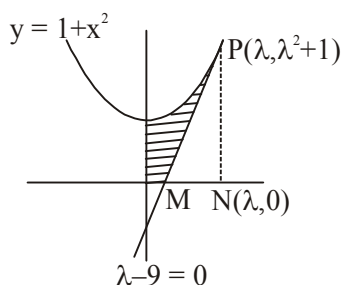
$$\text{Area} \int_0^1 (4e^{x/4} - e^{x/4}) dx = 12(e^{1/4} - 1)$$

I04/2/017. A tangent line with positive slope to a parabola $y = x^2 + 1$ is drawn such that the area of the figure bounded by the parabola, the tangent line and the coordinate axes is $\frac{11}{3}$. Given that the y-intercept of the tangent is negative, then the x-coordinate of the point of tangency is greater than :

- (A) 1 (B) 2 (C) $\frac{7}{3}$ (D) 3

Ans. ABC

Sol.



Equation of tangent at $(\lambda, \lambda^2 + 1)$ is $y - (\lambda^2 + 1) = 2\lambda(x - \lambda)$

$$\Rightarrow y = 2\lambda x - \lambda^2 + 1$$

$$\therefore \text{Required area} = \int_0^{\lambda} (x^2 + 1) dx - \text{ar } (\triangle PMN)$$

$$\Rightarrow \frac{\lambda^3}{3} + \lambda - \frac{1}{2} \frac{(\lambda^2 + 1)^2}{2\lambda} = \frac{11}{3}$$

$$\Rightarrow 3\lambda^4 + 18\lambda^2 - 132\lambda - 9 = 0$$

$$\therefore (\lambda - 3)(3\lambda^3 + 9\lambda^2 + 45\lambda + 3) = 0$$

↓

not satisfy by any positive value of λ .

$$\lambda = 3$$

I04/2/018. Consider $f(x) = x^2 - 4x + 3 \forall x \in \mathbb{R}$. Identify correct option(s) :

(A) Area bounded by curve $y = f(|x|)$ and x-axis, above the x-axis is $\frac{10}{3}$ sq. units.

(B) Area bounded by curve $y = |f(|x|)|$ between $1 \leq |x| \leq 3$ and x-axis is $\frac{8}{3}$ sq. units.

(C) Number of solutions of equation $|f(x)| = 1$ is 3

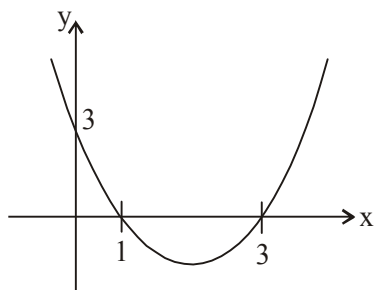
(D) Equation $|f(|x|)| = a$ has exactly 4 roots for $a \in (1, 3)$ only.

Ans.

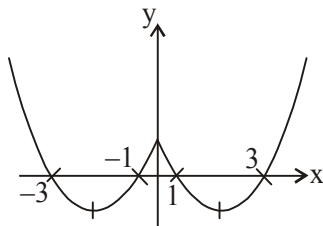
BC

Sol.

(A) Graph of $f(x)$



Graph of $y = f(|x|)$:

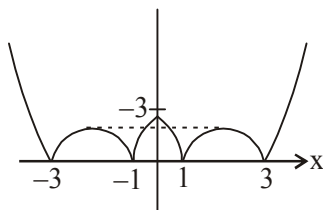


Required area

$$= 2 \cdot \int_0^1 (x^2 - 4x + 3) dx = 2 \left(\frac{x^3}{3} - \frac{4x^2}{2} + 3x \right) \Bigg|_0^1$$

$$= \frac{8}{3} \text{ sq. units}$$

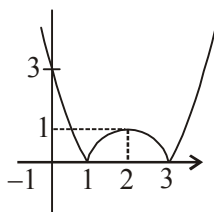
(B) Graph of $y = |f(|x|)|$



Required area

$$= \left| 2 \cdot \int_1^3 (x^2 - 4x + 3) dx \right| = \frac{8}{3} \text{ sq. units}$$

(C) Graph of $y = |f(x)|$



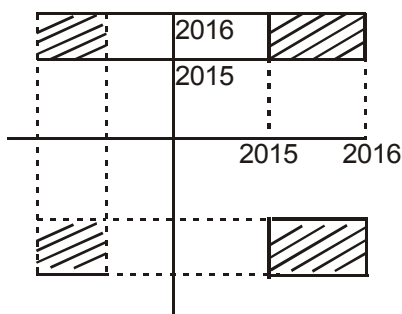
$\therefore$ 3 solution

I04/2/019. The area bounded by $\min(|x|, |y|) = 2015$ and $\max(|x|, |y|) = 2016$ is less than or equal to :

(A) 4 sq unit (B) 8 sq unit (C) 2 sq unit (D) 1 sq unit

Ans. AB

Sol. Required area $= 4 \times (1 \times 1) = 4$ sq unit



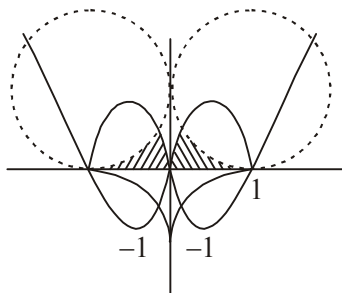
I04/2/020. The area of the region bounded between the curves $y = e^{|x|} \ln |x|$, $x^2 + y^2 - 2(|x| + |y|) + 1 \geq 0$ and x-axis where $|x| \leq 1$, if α is the x-coordinate of the point of intersection of curves in 1st quadrant, is :

- (A) $4 \left[\int_0^\alpha e^x \ln x dx + \int_\alpha^1 \left(1 - \sqrt{1 - (x-1)^2} \right) dx \right]$ (B) $4 \left[\int_0^\alpha e^x \ln x dx - \int_0^\alpha \left(1 - \sqrt{1 - (x-1)^2} \right) dx \right]$
 (C) $2 \left[-\int_0^\alpha e^x \ln x + \int_\alpha^1 \left(1 - \sqrt{1 - (x-1)^2} \right) dx \right]$ (D) $2 \left[\int_0^\alpha e^x \ln x + \int_\alpha^1 \left(1 - \sqrt{1 - (x-1)^2} \right) dx \right]$

Ans. C

Sol. Required area is

$$2 \left[\int_0^\alpha e^x \ln x + \int_\alpha^1 \left(1 - \sqrt{1 - (x-1)^2} \right) dx \right]$$



I04/2/021. A curve $y = 2x - 3x^3$ is intersected by a line $y = c$ at points $A(a, c)$ and $B(b, c)$, $b > a$, in the first quadrant. If the area bounded by y-axis, the curve and the line $y = c$ is equal to area bounded by the line $y = c$ and the curve (both the regions are in first quadrant), then :

- (A) $c = \frac{2}{3}$ (B) $b = \frac{2}{3}$ (C) $\int_0^b (2x - 3x^3) dx = bc$ (D) $c = \frac{4}{9}$

Ans. ABD

Sol.

$$\because A_1 = A_2 \Rightarrow \int_0^a (c - (2x - 3x^3)) dx = \int_a^b ((2x - 3x^3) - c) dx$$

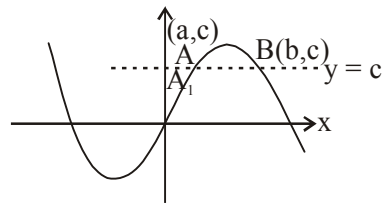
$$\Rightarrow b^2 - \frac{3}{4}b^4 = bc \quad \dots(1)$$

$$\text{Also } 2b - 3b^3 = c \quad \dots(2)$$

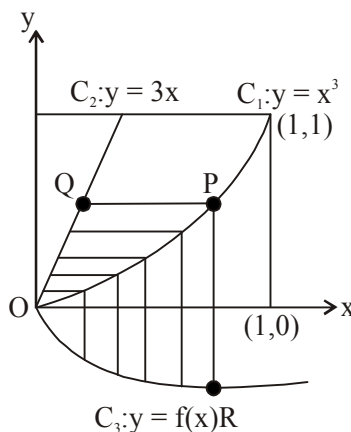
$$\{\because y = 2x - 3x^3\}$$

from equations (1) and (2)

$$\Rightarrow b = \frac{2}{3}, c = \frac{4}{9}$$



I04/2/022. Let C_1 and C_2 be the graphs of $y = x^3$ and $y = 3x$, $x \in [0, 1]$ respectively. Let C_3 be the graph of a function $y = f(x)$ in $[0, 1]$, $f(0) = 0$. For a point P on C_1 , let the lines through P , parallel to the axes meet C_2 and C_3 at Q and R respectively (see figure). If $f(x)$ is such that for every position of P (on C_1) the areas of shaded regions OPQ & ORP are equal, then :



$$(A) \int_0^1 f(x) dx = -\frac{1}{3} \quad (B) f\left(\frac{1}{2}\right) = -\frac{7}{32} \quad (C) f\left(\frac{1}{2}\right) = -\frac{5}{32} \quad (D) f'(x) < 0 \forall x \in (0, 1)$$

Ans. ABD

Sol. Let $P(\alpha, \alpha^3)$

$$\int_0^\alpha (x^3 - f(x)) dx = \int_0^{\alpha^3} \left(y^{1/3} - \frac{y}{3} \right) dy$$

$$\alpha^3 - f(\alpha) = \left(\alpha - \frac{\alpha^3}{3} \right) 3\alpha^2$$

$$f(\alpha) = \alpha^5 - 2\alpha - 2\alpha^3 \Rightarrow f(x) = x^5 - 2x^3$$

$$(A) \int_0^1 (x^2 - 2x^3) dx = \frac{1}{6} - \frac{1}{2} = -\frac{1}{3}$$

$$(B) f\left(\frac{1}{2}\right) = \frac{1}{32} - \frac{1}{4} = -\frac{7}{32}$$

$$(D) f(x) = x^2(5x^2 - 6) < 0 \quad \forall x \in (0, 1)$$

I04/2/023. If A denotes the area of the curve $2\{y\} = [x] + 1$ for $0 \leq y < 1$ and $0 \leq x \leq 1$, then A is less than or equal to (where $\{.\}$ and $[.]$ are the fractional part and greatest integer functions respectively) :

$$(A) \frac{\pi}{6}$$

$$(B) \frac{3}{4}$$

$$(C) \frac{1}{2}$$

$$(D) \frac{1}{3}$$

Ans. ABC

Sol. $2\{y\} = [x] + 1 \quad \dots(1)$

$$\because 0 \leq y < 1 \quad \Rightarrow \{y\} = y$$

$$\therefore (1) \Rightarrow 2y = [x] + 1$$

$$0 \leq [x] + 1 < 2$$

$$-1 \leq [x] < 1$$

but $[x]$ is an integer

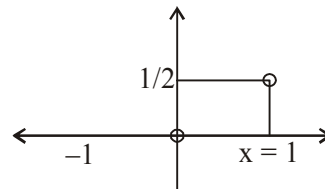
$$\Rightarrow [x] = -1 \quad \Rightarrow -1 \leq x < 0$$

$$[x] = 0 \quad \Rightarrow 0 \leq x < 1$$

$$\text{When } x \in [-1, 0) \quad \Rightarrow y = 0$$

$$\text{when } x \in [0, 1) \quad \Rightarrow y = \frac{1}{2}$$

$$\therefore \text{Area} = \frac{1}{2}$$



I02/2/024. If $f(x) = e^x + \int_0^1 (e^x + te^{-x}) f(t) dt$ then :

$$(A) f(0) = \frac{e^2 - e + 2}{4e - 2e^2}$$

$$(B) f(1) = \frac{2e^2 + e - 2}{4e - 2e^2}$$

$$(C) f(0) = \frac{e^2 + e - 2}{4e - 2e^2}$$

$$(D) f(0) = \frac{2e^2 - e - 1}{4e - 2e^2}$$

Ans. CD

Sol. $f(x) = e^x \left(1 + \int_0^1 f(t) dt \right) + e^{-x} \int_0^1 t f(t) dt$

Let $A = 1 + \int_0^1 f(t) dt$ & $B = \int_0^1 t f(t) dt$

So; $f(x) = e^x(A) + Be^{-x}$

Now; $A = 1 + \int_0^1 (Ae^t + Be^{-t}) dt$

$$A = 1 + A(e-1) + B \left(1 - \left(\frac{1}{e} \right) \right)$$

$$(2-e)A + \left(\left(\frac{1}{e} \right) - 1 \right) B = 1 \quad \dots(i)$$

Now; $B = \int_0^1 t(Ae^t + Be^{-t}) dt$

$$B = A + B \left(1 - \frac{2}{e} \right)$$

$$A - 2e^{-1}B = 0 \quad \dots(ii)$$

From (i) and (ii); $A = \frac{2(e-1)}{4e-2e^2}; B = \frac{e-1}{4-2e}$

Hence, $f(x) = \frac{2(e-1)}{4e-2e^2} \cdot e^x + \frac{e-1}{4-2e} e^{-x}$

$$f(0) = \frac{2(e-1)}{4e-2e^2} + \frac{e-1}{4-2e} = \frac{2e-2+e^2-e}{4e-2e^2} = \frac{e^2+e-2}{4e-2e^2}$$

$$f(1) = \frac{2(e-1)}{4e-2e^2} e + \frac{e-1}{4-2e} (e^{-1}) = \frac{2(e^2-e)}{4e-2e^2} + \frac{e-1}{4e-2e^2}$$

$$f(1) = \frac{2e^2-e-1}{4e-2e^2}$$

I02/2/025. A real valued function $f(x)$ which is monotonic and differentiable.

If $\int_{f(a)}^{f(b)} 2x \{b - f^{-1}(x)\} dx = \int_a^b \{g(x) - (f(a))^2\} dx$ then :

- (A) min value of $g(x)$ is 4 if $f(x) = x + \frac{1}{x}$
 (B) max. value of $g(x)$ is 2 if $f(x) = \sin x + \cos x$
 (C) range of $g(x)$ is $[4, \infty]$ if $f(x) = e^x + e^{-x}$
 (D) max. value of $g(x)$ is 4 if $f(x) = x + \frac{1}{x}$

Ans. ABC

Sol. Let
$$I = \int_{f(a)}^{f(b)} 2x \{b - f^{-1}(x)\} dx$$

Let $f^{-1}(x) = t \Rightarrow x = f(t)$
 $dx = f'(t)dt$

Now,
$$I = \int_a^b \underbrace{(b-t)}_I \underbrace{(2f(t)f'(t))}_{II} dt$$

$$I = \left[(b-t)(f(t))^2 \right]_a^b - \int_a^b (-1)(f(t))^2 dt$$

$$I = \int_a^b (f(t))^2 dt - (b-a)(f(a))^2 = \int_a^b (f(t))^2 - (f(a))^2 dt$$

So,
$$I = \int_a^b \left\{ (f(x))^2 - (f(a))^2 \right\} dx$$

$$g(x) = (f(x))^2$$

Now, if $f(x) = x + \frac{1}{x}$; $(f(x))^2 = x^2 + \frac{1}{x^2} + 2$

$$g(x) = (f(x))^2 = x^2 + \frac{1}{x^2} + 2$$

Range of $(g(x))$ is $[4, \infty)$

$$(g(x))_{\min} = 4 \text{ (if } f(x) = x + \frac{1}{x} \text{)}$$

Now, if $f(x) = \sin x + \cos x$

then $g(x) = (\sin x + \cos x)^2 = 1 + \sin 2x$

$$(g(x))_{\max} = 2$$

If $f(x) = e^x + e^{-x}$

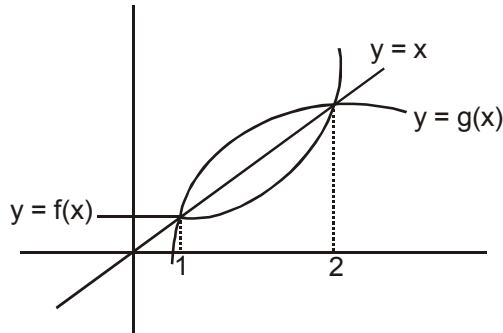
Then $g(x) = (f(x))^2 = (e^x + e^{-x})^2 = e^{2x} + e^{-2x} + 2 \Rightarrow (f(x))_{\min} = 4$

I02/2/026. Let $y = f(x)$ be a twice differentiable function such that $f'(x) > 0 \forall x \in \mathbb{R}$. Let $g(x) = f^{-1}(x)$ ($f(x)$ is invertible). $f(x) = g(x)$ has two distinct real roots $x = 1$ and 2 only. If $\int_1^2 f(x) dx = 1$ then which of the following must be correct ?

- (A) $g''(x) < 0 \forall x \in \mathbb{R}$ (B) $\int_1^2 g(x) dx = 2$ (C) $g'(x) < 0 \forall x \in \mathbb{R}$ (D) $f'(x) > 0 \forall x \in \mathbb{R}$

Ans. ABD

Sol.



$y = f(x)$ is increasing

$$\int_1^2 (x - f(x)) dx = \frac{1}{2} \text{ and } \int_1^2 g(x) dx = 2$$

I02/2/027. $\int_0^1 \frac{(2 + \sqrt{x}) dx}{(x + \sqrt{x} + 1)^2}$ is :

- (A) Irrational (B) Rational (C) Less than one (D) Greater than 2

Ans. BC

Sol. $\int_0^1 \frac{\left(\frac{2}{x^2} + \frac{1}{x^{3/2}}\right) dx}{\left(1 + \frac{1}{\sqrt{x}} + \frac{1}{x}\right)^2}$ Let $1 + \frac{1}{\sqrt{x}} + \frac{1}{x} = t$

$$-\int_{\infty}^3 \frac{2dt}{t^2} = 2 \int_3^{\infty} \frac{dt}{t^2} \quad \left(-\frac{1}{2x^{3/2}} - \frac{1}{x^2}\right) dx = dt$$

$$= 2 \left(-\frac{1}{2t}\right)_3^{\infty} \quad \left(\frac{1}{x^{3/2}} + \frac{2}{x^2}\right) dx = -2dt$$

$$= -\left(\frac{1}{t}\right)_3^\infty = \frac{1}{3}$$

I02/2/028. If $f: (0, 1) \rightarrow \mathbb{R}$ is defined by $f(x) = \sum_{n=0}^{\infty} \frac{x^{2^n}}{1-x^{2^{n+1}}}$, then :

(A) $f^{-1}(x) = \frac{x}{x+1}$

(B) $\int_0^{\frac{1}{2}} f(x) dx = \ln 2 - \frac{1}{2}$

(C) $\int_0^1 f(x) dx = \ln 2 - 1$

(D) $\int_0^1 f(x) dx = \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots \dots \infty$

Ans. ABD

Sol. $f(x) = \sum_{n=0}^{\infty} \frac{1}{1-x^{2^n}} - \frac{1}{1-x^{2^{n+1}}} = \frac{1}{1-x} - 1 = \frac{x}{1-x}$

I02/2/029. If $\frac{\pi}{2} < \alpha < \frac{2\pi}{3}$ and $I = \int_0^{\sin 2\alpha} \frac{dx}{\sqrt{4\cos^2 \alpha - x^2}}$ then $[2I]$ may be ([.] is GIF) :

(A) -1

(B) -4

(C) -3

(D) None

Ans. BC

Sol. $I = \left(\sin^{-1} \left(\frac{x}{2\cos \alpha} \right) \right)_0^{\sin 2\alpha} = \sin^{-1}(\sin \alpha) = \pi - \alpha ;$

$I = \alpha - \pi$

$\therefore 2I \in \left(-\frac{2\pi}{3}, -\pi \right) \left(-\frac{2\pi}{3}, -\pi \right)$

I02/2/030. The value of the definite integral $\int_{-\infty}^a \frac{(\sin^{-1} e^x + \sec^{-1} e^{-x}) dx}{(\tan^{-1} e^a + \tan^{-1} e^x)(e^x + e^{-x})}$ ($a \in \mathbb{R}$) is :

(A) Independent of a

(B) dependent on a

(C) $\frac{\pi}{2} \ln 2$

(D) $\frac{\pi}{2} \ln(2 \tan^{-1} e^a)$

Ans. AC

Sol. We have $I = \int_{-\infty}^a \left(\frac{\sin^{-1} e^x + \cos^{-1} e^x}{\tan^{-1} e^a + \tan^{-1} e^x} \right) \left(\frac{e^x}{e^{2x} + 1} \right) dx = \frac{\pi}{2} \int_{-\infty}^a \frac{1}{(\tan^{-1} e^a + \tan^{-1} e^x)} \left(\frac{e^x}{(e^{2x} + 1)} \right) dx$

Put $\tan^{-1} e^x = t \Rightarrow \frac{e^x}{e^{2x} + 1} dx = dt$

$$I = \frac{\pi}{2} \int_0^{\tan^{-1} e^a} \frac{dt}{(t + \tan^{-1} e^a)} = \frac{\pi}{2} \left[\ln(t + \tan^{-1} e^a) \right]_0^{\tan^{-1} e^a} = \frac{\pi}{2} \left[\ln(2 \tan^{-1} e^a) - \ln(\tan^{-1} e^a) \right] = \frac{\pi}{2} \ln 2$$

I02/2/031. Let $f(x) = \int_1^x \frac{3^t}{1+t^2} dt$, $x > 0$ then :

(A) for $0 < \alpha < \beta$, $f(\alpha) < f(\beta)$

(B) for $0 < \alpha < \beta$, $f(\alpha) > f(\beta)$

(C) for all $x > 0$, $f(x) + \frac{\pi}{4} < \tan^{-1} x$

(D) for all $x > 0$, $f(x) + \frac{\pi}{4} > \tan^{-1} x$

Ans. A

Sol. We have $f'(x) = \frac{3^x}{1+x^2} > 0 \forall x > 0$

$$\therefore f'(x) = \frac{3^x}{1+x^2} > \frac{1}{1+x^2} > 0 \forall x > 0$$

$$\Rightarrow \int_1^x f'(x) dx > \int_1^x \frac{1}{1+x^2} dx$$

$$\Rightarrow f(x) > \tan^{-1} x - \tan^{-1} 1$$

$$\text{Sp, } f(x) + \frac{\pi}{4} > \tan^{-1} x, \text{ for all } x > 0$$

Note that in (D) equality holds at $x = 1$ and hence (d) is false

I02/2/032. If $A = \int_0^{\infty} \frac{\sin x}{x} dx$, then the value of the integral $\int_0^{\infty} \frac{\sin ax \cos bx}{x} dx$ must be :

(A) A if $a > b > 0$

(B) 2A if $a + b = 0$

(C) $\frac{A}{a+b}$

(D) 0 if $b > a > 0$

Ans. AD

Sol.

$$\text{Let } I = \frac{1}{2} \int_0^{\infty} \frac{\sin(a+b)x + \sin(a-b)x}{x} dx$$

$$\Rightarrow I = \frac{1}{2} \int_0^{\infty} \frac{\sin(a+b)x}{(a+b)x} (a+b) dx + \frac{1}{2} \int_0^{\infty} \frac{\sin(a-b)x}{(a-b)x} (a-b) dx$$

$$\Rightarrow I = \begin{cases} \frac{1}{2} A + \frac{1}{2} A = A & \text{if } a > b > 0 \\ \frac{1}{2} A - \frac{1}{2} A = 0 & \text{if } 0 < a < b \end{cases}$$

I02/2/033. If $f(x) = \sqrt{1-x^2} (\cos^2(\tan^{-1} x) - x^2 \sin^2(\tan^{-1} x))$, then :

(A) Range of $f(x)$ is $\mathbb{R}$

(B) Range of $f(x)$ is $[0, 1]$

(C) $\int_0^1 f(x) dx$ is $\frac{3\pi}{16}$

(D) $\int_0^1 f(x) dx$ is $\frac{3}{8}$

Ans.

BC

Sol.

$$f(x) = \sqrt{1-x^2} \left(\frac{1}{1+x^2} - x^2 \cdot \frac{x^2}{1+x^2} \right)$$

$$= \sqrt{1-x^2} (1-x^2) = (1-x^2)^{3/2}$$

Range of $f(x)$ is $[0, 1]$

$$\text{Now } \int_0^1 (1-x^2)^{3/2} dx = \int_0^{\pi/2} \cos^4 \theta d\theta$$

I02/2/034. If $P(\alpha, \beta) = \int_{\sin(\sin^{-1} \alpha)}^{\cos(\cos^{-1} \beta)} \frac{|\cos(\cos^{-1} x)|}{|\sin(\sin^{-1} x)|} dx$, then :

(A) maximum value of $P(\alpha, \beta)$ is 2

(B) minimum value of $P(\alpha, \beta)$ is -2

(C) $P(\alpha, \beta) = P(\beta, \alpha)$

(D) $P(\alpha, \beta) + P(\beta, \alpha) = 0$

Ans.

ABD

Sol.

$$P(\alpha, \beta) = \int_{\alpha}^{\beta} dx = \beta - \alpha$$

maximum value of $P(\alpha, \beta)$ is 2

as $-1 \leq \alpha \leq 1, -1 \leq \beta \leq 1$

minimum value of $P(\alpha, \beta)$ is 2

$$P(\alpha, \beta) = \beta - \alpha$$

$$P(\beta, \alpha) = \alpha - \beta$$

$$\Rightarrow P(\alpha, \beta) + P(\beta, \alpha) = 0$$

I02/2/035. The value of $I = \int_{-1}^1 \left(A(\sin^{-1} x) + B \left(\frac{e^{2x} - 1}{e^{2x} + 1} \right) + C[x^2] + D \sin |x| \right) dx$ is independent of {where A, B, C,

$D \in \mathbb{R}$ and $[.]$ denotes greatest integer function} :

(A) A

(B) B

(C) C

(D) D

Ans. ABC

Sol. $\sin^{-1} x, \frac{e^{2x} - 1}{e^{2x} + 1}$ are odd functions

and $x^2 \in [0, 1) \forall x \in (-1, 1)$

I02/2/036. Let $f(x)$ is an even function such that $f(x) + f(x-3) = x(x-3) + 1$, if $\int_0^3 \frac{f(x) dx}{(x^2 - 3x + 1)} = \frac{a}{b}$ where a, b are

prime number, then $a + b$ is :

(A) prime number

(B) odd

(C) even

(D) composite

Ans. AB

Sol. Let $I = \int_0^3 \frac{f(x) dx}{(x^2 - 3x + 1)}$ (1)

$$I = \int_0^3 \frac{f(3-x)}{(x^2 - 3x + 1)} \dots\dots(2)$$

$$(1) + (2) \Rightarrow 2I = 3$$

$$I = \frac{3}{2}$$

I02/2/037. For a positive integer n, let $I_n = \int_{-\pi}^{\pi} \left(\frac{\pi}{2} - |x| \right) \cos nx \, dx$. Then value of $9(I_1 + I_2 + I_3 + I_4)$:

(A) even number

(B) odd number

(C) composite number

(D) Prime number

Ans. AC

Sol. We have, $I_n = \int_{-\pi}^{\pi} \left(\frac{\pi}{2} - |x| \right) \cos nx \, dx = 2 \int_0^{\pi} \left(\frac{\pi}{2} - x \right) \cos nx \, dx = 2 \int_0^{\pi} \left(\frac{\pi}{2} - x \right) \cos nx \, dx$

$$= \pi \int_0^{\pi} \cos nx \, dx - 2 \int_0^{\pi} x \cos nx \, dx = \pi \left[\frac{\sin nx}{n} \right]_0^{\pi} - 2 \int_0^{\pi} x \cos nx \, dx = -2 \left[\frac{x \sin nx}{n} \right]_0^{\pi} + 2 \int_0^{\pi} \frac{\sin nx}{n} \, dx$$

$$= \frac{2}{n} \left[\frac{-\cos nx}{n} \right]_0^{\pi} = \frac{2(1 - \cos n\pi)}{n^2} = \begin{cases} \frac{4}{n^2}, & \text{if } n \text{ is odd} \\ 0, & \text{if } n \text{ is even} \end{cases}$$

Hence, $I_1 + I_2 + I_3 + I_4 = \frac{4}{1^2} + \frac{4}{3^2} = \frac{40}{9}$

Hence, $9(I_1 + I_2 + I_3 + I_4) = \frac{9 \times 40}{9} = 40$

I02/2/038. The minimum value of the function $f(x) = \int_0^x \frac{d\theta}{\cos \theta} + \int_x^{\pi/2} \frac{d\theta}{\sin \theta}$ where $x \in \left(0, \frac{\pi}{2}\right)$, is :

(A) $2 \ln(\sqrt{2} + 1)$ (B) $-2 \ln(\sqrt{2} - 1)$ (C) $\ln(2 + \sqrt{3})$ (D) $-\ln(2 - \sqrt{3})$

Ans. AB

Sol. $f(x) = \int_0^x \frac{d\theta}{\cos \theta} + \int_x^{\pi/2} \frac{d\theta}{\sin \theta}$

$$\Rightarrow f'(x) = \frac{1}{\cos x} - \frac{1}{\sin x}$$

$$= \frac{\sin x - \cos x}{\sin x \cos x}$$

$f'(x)$ change sign $-$ to $+$ at $x = \frac{\pi}{4}$

so $f(x)$ takes minimum value at $x = \frac{\pi}{4}$

$$f(x) = \int_0^{\pi/4} \sec \theta \, d\theta + \int_{\pi/4}^{\pi/2} \cos \sec \theta \, d\theta$$

$$= (\ln(\sec \theta + \tan \theta))_0^{\pi/4} + (\ln(\operatorname{cosec} \theta - \cot \theta))_{\pi/4}^{\pi/2}$$

$$= \ln(\sqrt{2} + 1) - \ln(1 + 0) + \ln 1 - \ln(\sqrt{2} - 1)$$

$$= \ln \left(\frac{\sqrt{2}+1}{\sqrt{2}-1} \right) = \ln (\sqrt{2}+1)^2 = 2 \ln (\sqrt{2}+1)$$

I02/2/039. If $L = \lim_{x \rightarrow 0} \int_0^x \frac{(\sin^2 t) \sqrt{a^2 - \sin^2 t}}{ax - b \tan x} dt$, where L is finite, then :

(A) $L \neq 0$ for $a \neq b$

(B) $L = 0$ for $a \neq b$

(C) $L = 1$ for $a > 0, b > 0$

(D) $L = -1$ for $a = b > 0$

Ans. BD

Sol.
$$L = \lim_{x \rightarrow 0} \frac{\int_0^x \sin^2 t \sqrt{a^2 - \sin^2 t} dt}{ax - b \tan x} \left(\frac{0}{0} \right)$$

$$= \lim_{x \rightarrow 0} \frac{\sin^2 x \sqrt{a^2 - \sin^2 x}}{a - b \sec^2 x} = \frac{0}{a - b}$$

$\Rightarrow$ if $a \neq b$, then $L = 0$

if $a = b$, then
$$L = \lim_{x \rightarrow 0} \frac{\sin^2 x \sqrt{a^2 - \sin^2 x}}{a(1 - \sec^2 x)}$$

$$= \lim_{x \rightarrow 0} \frac{-\cos^2 x \sqrt{a^2 - \sin^2 x}}{a} = \frac{|a|}{a} = -1 \text{ for } a > 0$$

I02/2/040. Let $f(x) = \int_1^x \frac{dt}{2+t^4}$, $f(2)$ satisfies inequality :

(A) $0 < f(2) < \frac{1}{2}$

(B) $\frac{1}{18} < f(2) < \frac{1}{3}$

(C) $f(2) > \frac{1}{3}$

(D) $f(2) > 1$

Ans. AB

Sol.
$$f'(x) = \frac{1}{2+x^4}$$

By LMVT $f'(c) = \frac{f(2)-f(1)}{2-1}$ for some $c \in (1, 2)$

$$\Rightarrow f(2) = \frac{1}{2+c^4} \text{ as } f(1) = 0 \Rightarrow 1 < c < 2$$

$$\Rightarrow 3 < 2+c^4 < 18 \Rightarrow f(2) < \frac{1}{3}$$

I02/2/041. If $(x + 1)(x + 2)(x + 3) \dots (x + 99) = A_0 + A_1x + A_2x^2 + \dots + A_{99}x^{99}$ and $\frac{\sum_{r=1}^{99} rA_r}{(100)!} = \sum_{i=1}^{99} \frac{1}{r_i}$, where

$r_i < r_j \forall i < j$, then the value of $\left[\sum_{k=1}^{99} \frac{1}{\sqrt{r_k - 1}} \right]$ is greater than or equal to (where $[.]$ denote greatest integer

function):

(A) 16

(B) 17

(C) 18

(D) 19

Ans. ABC

Sol. $\therefore f(x) = (x + 1)(x + 2) \dots (x + 99) = A_0 + A_1x + A_2x^2 + \dots + A_{99}x^{99}$

$$\Rightarrow \sum_{r=1}^{99} r \cdot A_r = f'(1) = (100)! \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{100} \right)$$

$$\Rightarrow r_k = k + 1$$

$$\therefore \left[\sum_{k=1}^{99} \frac{1}{\sqrt{r_k - 1}} \right] = \left[\sum_{k=1}^{99} \frac{1}{\sqrt{k}} \right] \geq \left[\int_1^{100} \frac{dx}{\sqrt{x}} \right] = 18$$

I02/2/042. Let $n \in \mathbb{N}$, $n > 11$, then $1 \cdot 3 \cdot 5 \cdot 7 \dots (2n - 1)$ is :

(A) Less than $\left(\frac{2n-1}{e} \right)^{\frac{2n-1}{2}}$

(B) Less than $\left(\frac{2n+1}{e} \right)^{\frac{2n+1}{2}}$

(C) Greater than $\left(\frac{2n-1}{e} \right)^{\frac{2n-1}{2}}$

(D) Greater than $\left(\frac{2n+1}{e} \right)^{\frac{2n+1}{2}}$

Ans. BC

Sol. Let $\lambda = 1 \cdot 3 \cdot 5 \cdot 7 \dots (2n - 1)$

$$\Rightarrow \ln \lambda = \ln 3 + \ln 5 + \ln 7 + \dots \ln (2n - 1)$$

$$\Rightarrow \ln \lambda < \frac{1}{2} \int_3^{2n+1} \ln x \, dx$$

$$\text{Also } \ln \lambda > \frac{1}{2} \int_1^{2n-1} \ln x \, dx$$

I02/2/043. Let 'f' be a derivable function such that $1 < f'(x) < 2 \forall x \in [0,1]$ & $f(0) = 1$, then $\int_0^1 f(x) dx$ is greater than :

- (A) 1 (B) $\frac{3}{2}$ (C) 2 (D) 3

Ans. AB

Sol. $1 < \frac{f(x) - f(0)}{x} < 2 < 2$

$$\int_0^1 (1+x) dx < \int_0^1 f(x) dx < \int_0^1 (1+2x) dx$$

$$\frac{3}{2} < f(x) < 2$$

I02/2/044. If $I_1 = \int_1^e \frac{\ln(x+1)}{x^2} dx$, $I_2 = \int_0^1 \frac{e^x}{(x+1)^2} dx$ then :

- (A) $I_1 > 1$ (B) $I_1 < 1$ (C) $I_2 < \ln 2$ (D) $I_1 > 1, I_2 > \ln 2$

Ans. BD

Sol. $\therefore \ln(x+1) < x$

$$\Rightarrow \frac{\ln(x+1)}{x^2} < \frac{1}{x} \Rightarrow I_1 < 1$$

$$\text{Now } x+1 < e^x \Rightarrow \frac{e^x}{(x+1)^2} > \frac{1}{x+1} \Rightarrow I_2 > \ln 2$$

I02/2/045. If $I = \int_0^1 \frac{\sqrt{\sin x}}{1+x} dx$, then :

- (A) $I < \frac{\pi}{4}$ (B) $I > \frac{\pi}{4}$ (C) $I < \frac{1-\cos 1}{2}$ (D) $I > \frac{1-\cos 1}{2}$

Ans. AD

Sol. As $\frac{\sin x}{2} < \frac{\sqrt{\sin x}}{1+x} < \frac{1}{1+x^2}$

$$\Rightarrow \frac{1}{2} \int_0^1 \sin x dx < I < \int_0^1 \frac{1}{1+x^2} dx$$

$$\Rightarrow \frac{1}{2}(1 - \cos 1) < 1 < \frac{\pi}{4}$$

- I02/2/046. Suppose that $f(x)$ is a differentiable invertible function with $f'(x) \neq 0$ and $h(x) = \int_1^x f(t) dt$. Given that $f(1) = f'(1) = 1$ and $g(x)$ is inverse of $f(x)$. Let $G(x) = x^2 g(x) - xh(g(x)) \forall x \in \mathbb{R}$. Which of the following are correct:
 (A) $G'(1) = 2$ (B) $G'(1) = 3$ (C) $G''(1) = 2$ (D) $G''(1) = 3$

Ans. AD

Sol. $h'(x) = f(x)$

$$h''(x) = f'(x)$$

$$h(1) = 0, f(1) = f'(1) = h'(1) = h''(1) = 1 = g(1)$$

$$f(g(x)) = x$$

$$\Rightarrow f'(g(x)) \cdot g'(x) = 1$$

$$\Rightarrow f'(g(1)) \cdot g'(1) = 1 \Rightarrow f'(1) \cdot g'(1) = 1 \Rightarrow g'(1) = 1$$

$$G(x) = x^2 g(x) - xh(g(x))$$

$$G'(x) = 2x g(x) + x^2 g'(x) - h(g(x)) - xh'(g(x)) \cdot g'(x)$$

$$= 2xg(x) + x^2 g'(x) - h(g(x)) - x^2 g'(x) = 2xg(x) - h(g(x))$$

$$G''(x) = 2g(x) + 2xg'(x) - h'(g(x)) \cdot g'(x) = 2g(x) + xg'(x)$$

$$G'(1) = 2g(1) - h(g(1)) = 2$$

$$g''(1) = 2g(1) + g'(1) = 3$$

- I02/2/047. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a real valued differentiable function satisfying the integral equation

$$1 + \int_1^x 2t^2 f(t) dt = x^2 + 2 \int_x^1 f(t) dt, \text{ then which of the following is (are) true :}$$

(A) $f(x)$ has exactly one asymptote

(B) $f(x)$ has neither injective nor surjective

(C) Distance between two horizontal tangents of $f(x)$ is 1

(D) If the equation $f(|x|) = k$ has four distinct solutions, then range of k is equal to $\left(0, \frac{1}{2}\right]$

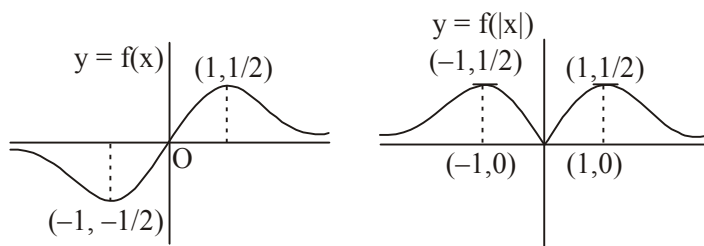
Ans. ABC

Sol.

Given equation $1 + \int_1^x 2t^2 f(t) dt = x^2 + 2 \int_x^1 f(t) dt$

diff. w.r.t x

$$2x^2 f(x) = 2x + (0 - 2f(x)) \Rightarrow f(x) = \frac{x}{1+x^2}$$



clearly $y = \frac{1}{2}$ is only the horizontal tangent of $f(|x|)$

and $f(|x|) = k$ has four solutions for $k \in \left(0, \frac{1}{2}\right)$

I02/2/048. Let $f(x)$ be a continuous function and c is a constant satisfying $\int_0^x f(t) dt = e^x - ce^{2x} \int_0^1 f(t) e^{-t} dt$ then :

(A) $f(x) = e^x + 2e^x$ (B) $f(x) = e^x - 2e^{2x}$ (C) $c = -\frac{1}{1+2e}$ (D) $c = \frac{1}{3-2e}$

Ans. BD

Sol. Put $x = 0$ in original equation

$$0 = 1 - c \int_0^1 f(t) e^{-t} dt = 1 - ck$$

$$c = \frac{1}{k} \text{ where } k = \int_0^1 f(t) e^{-t} dt \quad (kc = 1)$$

$$\text{differentiating } \int_0^x f(t) dt = e^x - e^{2x} - ck$$

the given equation, we get $f(x) = e^x - 2e^{2x} \Rightarrow$ (B)

$$\text{again } k = \int_0^1 (e^t - 2e^{2t}) e^{-t} dt = \int_0^1 (1 - 2e^t) dt = 3 - 2e$$

$$\therefore c = \frac{1}{k} = \frac{1}{3-2e} \Rightarrow \text{ (D)}$$

I02/2/049. Let $f(x) = \begin{cases} x+1 & 0 \leq x \leq 1 \\ 2x^2 - 6x + 6, & 1 < x \leq 2 \end{cases}$ and $g(t) = \int_{t-1}^1 f(x) dx$ for $t \in [1, 2]$. Which of the following hold(s)

good ?

(A) $f(x)$ is continuous and differentiable in $[0, 2]$

(B) $g'(t)$ vanishes for $t = 3/2$ and 2

(C) $g(t)$ is maximum at $t = 3/2$

(D) $g(t)$ is minimum at $t = 1$

Ans. BCD

Sol. (A) $f(1^+) = f(1^-) = 2 \Rightarrow f(x)$ is continuous.

$f'(1^+) \neq f'(1^-) \Rightarrow f(x)$ is not differentiable.

$$(B) g(t) = \int_{t-1}^1 f(x) dx = \int_{t-1}^1 (x+1) dx = \frac{1}{2}(4-t^2)$$

$$g'(t) = -t$$

I02/2/050. If $G(x, t) = \begin{cases} x(t-1), & \text{where } x \leq t \\ t(x-1), & \text{where } x > t \end{cases}$

and if t is continuous function of x in $[0, 1]$.

$$\text{let } g(x) = \int_0^1 f(t).G(x, t)dt \text{ then :}$$

$$\text{यदि } G(x, t) = \begin{cases} x(t-1), & \text{जहाँ } x \leq t \\ t(x-1), & \text{जहाँ } x > t \end{cases} \text{ है}$$

तथा t अन्तराल $[0, 1]$ में x में एक सतत फलन है

$$\text{माना } g(x) = \int_0^1 f(t).G(x, t)dt \text{ है तो :}$$

$$(A) g(0) = 0$$

$$(B) g(0) = 1$$

$$(C) g(1) = 0$$

$$(D) g''(x) = f(x)$$

Ans. ACD

Sol. $x = 0, G(0, t) = 0, t \geq 0$

$$g(0) = \int_0^1 f(t) G(0, t) dt = 0$$

$$x = 1, G(1, t) = 0, t < 1$$

$$g(1) = \int_0^1 f(t) G(1, t) dt = 0$$

$$g(x) = \int_0^x f(t) G(x, t) dt + \int_x^1 f(t) G(x, t) dt = \int_0^x f(t) t(x-1) dt + \int_x^1 f(t) x(t-1) dt$$

$$g''(x) = f(x)$$

I02/2/051. If $I_n = \int_0^1 (1-x^2)^n dx$ ($n = 0, 1, 2, \dots$) then which of the following is/are correct ?

यदि $I_n = \int_0^1 (1-x^2)^n dx$ ($n = 0, 1, 2, \dots$) तब निम्न में से कौनसा/से सत्य होगा/होंगे ?

$$(A) \frac{I_{n+1}}{I_n} = \frac{2n+2}{2n+3}$$

$$(B) I_n = \prod_{r=1}^n \left(\frac{2r}{2r+1} \right) (n \in \mathbb{N})$$

$$(C) \frac{I_{n+1}}{I_n} = \frac{2n}{2n+1}$$

$$(D) I_n = \prod_{r=1}^n \left(\frac{2r+2}{2r+3} \right) (n \in \mathbb{N})$$

Ans. AB

Sol.
$$I_n = \int_0^1 (1-x^2)^n dx = x(1-x^2)^n \Big|_0^1 + 2n \int_0^1 x^2 (1-x^2)^{n-1} dx = 2n \cdot \int_0^1 ((1-x^2)^{n-1} - (1-x^2)^n) dx$$

$$\Rightarrow I_n = 2nI_{n-1} - 2nI_n \Rightarrow \frac{I_n}{I_{n-1}} = \frac{2n}{2n+1}$$

I02/2/052. If $I_n = \int_0^{\pi/2} x(\sin x + \cos x)^n dx$:

यदि $I_n = \int_0^{\pi/2} x(\sin x + \cos x)^n dx$:

$$(A) \frac{101 I_{101} - \frac{\pi}{2}}{I_{99}} = 200$$

$$(B) \frac{4 I_4 - \frac{\pi}{2}}{I_2} = 6$$

$$(C) \frac{101 I_{101} - \frac{\pi}{2}}{I_{99}} = 6$$

$$(D) \frac{4 I_4 - \frac{\pi}{2}}{I_2} = 8$$

Ans. AB

Sol. $I_n = \int_0^{\frac{\pi}{2}} x(\sin x + \cos x)^n dx$

$$I_n = \int_0^{\frac{\pi}{2}} x \left(\frac{\pi}{2} - x \right) (\cos x + \sin x)^n dx \Rightarrow I_n = \frac{\pi}{4} \int_0^{\frac{\pi}{2}} (\cos x + \sin x)^n dx \Rightarrow (\sqrt{2})^n \frac{\pi}{4} \int_0^{\frac{\pi}{2}} \left[\sin \left(x + \frac{\pi}{4} \right) \right]^n dx$$

$$I_n = (\sqrt{2})^n \frac{\pi}{4} \int_0^{\frac{\pi}{2}} \left(\sin \left(x + \frac{\pi}{4} \right) \right)^{n-1} \sin \left(x + \frac{\pi}{4} \right) dx$$

$$I_n = (\sqrt{2})^n \frac{\pi}{4} \left[-\sin \left(x + \frac{\pi}{4} \right)^{n-1} \cos \left(x + \frac{\pi}{4} \right) \right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} (n-1) \left(\sin \left(x + \frac{\pi}{4} \right) \right)^{n-2} - \left(\sin \left(x + \frac{\pi}{4} \right) \right)^n dx$$

$$I_n = \frac{\pi}{2} + 2(n-1)I_{n-2} - (n-1)I_n$$

$$\frac{n I_n - \frac{\pi}{2}}{I_{n-2}} = (n-1) \cdot 2$$

I02/2/053. If $u_n = \int_0^{\pi/2} \cos^n x \cos n x dx$:

यदि $u_n = \int_0^{\pi/2} \cos^n x \cos n x dx$:

(A) $u_1 = \frac{\pi}{4}$

(B) $u_2 = \frac{\pi}{8}$

(C) $u_3 = \frac{\pi}{32}$

(D) $u_4 = \frac{\pi}{32}$

Ans. ABD

Sol. We have,

$$u_n = \int_0^{\pi/2} \cos^n x \cos n x dx$$

$$\Rightarrow u_n = \left[\frac{\cos^n x \sin n x}{n} \right]_0^{\pi/2} + n \int_0^{\pi/2} \cos^n x \sin x \frac{\sin n x}{n} dx$$

$$\Rightarrow u_n = \int_0^{\pi/2} \cos^{n-1} x \sin x \sin n x dx$$

$$\Rightarrow u_n = \int_0^{\pi/2} \cos^{n-1} x \{ \cos(n-1)x - \cos x \cos nx \} dx$$

$$[\because \cos(n-1)x = \cos nx \cos x + \sin nx \sin x]$$

$$\Rightarrow u_n = \int_0^{\pi/2} \cos^{n-1} x \cos(n-1)x dx - \int_0^{\pi/2} \cos^n x \cos nx dx$$

$$\Rightarrow u_n = u_{n-1} - u_n$$

$$\Rightarrow 2u_n = u_{n-1}$$

$$\Rightarrow \frac{u_n}{u_{n-1}} = \frac{1}{2} \text{ for all } n = 2, 3, \dots$$

$$\Rightarrow u_1, u_2, u_3, \dots \text{ from a G.P. with common ratio } \frac{1}{2}.$$

Now,

$$u_n = u_1 \left(\frac{1}{2} \right)^{n-1}$$

$$\Rightarrow u_n = \frac{\pi}{4} \left(\frac{1}{2} \right)^{n-1}$$

$$\Rightarrow u_n = \frac{\pi}{2^{n+1}} \quad \left[\because u_1 = \int_0^{\pi/2} \cos^2 x dx = \frac{\pi}{4} \right]$$

I02/2/054. Let $I_n = \int_0^{\pi/2} \frac{\sec^2 x}{(\sec x + \tan x)^n} dx$; $n > 1$ then which of the following statement is/are correct :

माना $I_n = \int_0^{\pi/2} \frac{\sec^2 x}{(\sec x + \tan x)^n} dx$; $n > 1$ तब निम्न में से कौनसा/से कथन सत्य होगा/होंगे ?

(A) $I_3 = \frac{3}{8}$

(B) $I_2 + I_4 = \frac{14}{15}$

(C) $I_3 + I_5 = \frac{5}{12}$

(D) $I_2 + I_3 = \frac{25}{24}$

Ans. ABD

Sol. $I_n = \int_0^{\pi/2} \sec x \left(\frac{\sec^2 x + \sec x \tan x}{(\sec x + \tan x)^{n+1}} \right) \cdot dx$

$$I_n = \left(\frac{\sec x}{-n(\sec x + \tan x)^n} \right)_0^{\pi/2} + \frac{1}{n} \int_0^{\pi/2} \frac{\sec x \cdot \tan x}{(\sec x + \tan x)^n} \cdot dx$$

$$= \frac{1}{n} + \frac{1}{n} \left[\int_0^{\pi/2} \frac{\sec x \tan x + \sec^2 x}{(\sec x + \tan x)^n} dx - \int_0^{\pi/2} \frac{\sec^2 x}{(\sec x + \tan x)^n} dx \right]$$

$$I_n = \frac{1}{n} + \frac{1}{n} \left[\frac{1}{(-n+1)(\sec x + \tan x)^{n-1}} \right]_0^{\pi/2} - \frac{1}{n} I_n \Rightarrow \left(1 + \frac{1}{n}\right) I_n = \frac{1}{n} + \frac{1}{n(n-1)} = \frac{1}{n-1} \Rightarrow I_n = \frac{n}{n^2-1}$$

I02/2/055. If $\int \cos(2016x)(\sin^{2014} x) dx = f(x) + c$, $f(0) = 0$ then :

- (A) $f(x)$ is odd (B) $\int_{\pi}^{2\pi} f(x) dx = 0$ (C) $\int_0^{\pi} f(x) dx = 0$ (D) $\lim_{x \rightarrow 0} \frac{2015f(x)}{x^{2015}} = 1$

यदि $\int \cos(2016x)(\sin^{2014} x) dx = f(x) + c$, $f(0) = 0$ है तब :

- (A) $f(x)$ विषम है (B) $\int_{\pi}^{2\pi} f(x) dx = 0$ (C) $\int_0^{\pi} f(x) dx = 0$ (D) $\lim_{x \rightarrow 0} \frac{2015f(x)}{x^{2015}} = 1$

Ans. ABCD

Sol.

I02/2/056. Choose the correct option(s), if

$$\int_{\frac{3}{2}}^x \frac{x^2 + 3}{x^4 + x^3 - x^2 - 3x + 9} dx = f(x) :$$

सही विकल्प/विकल्पों का चयन करें, यदि

$$\int_{\frac{3}{2}}^x \frac{x^2 + 3}{x^4 + x^3 - x^2 - 3x + 9} dx = f(x) :$$

- (A) $f(2) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{2}{\sqrt{19}}$ (B) $f(2) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{4}{\sqrt{19}}$
 (C) $f(3) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{5}{\sqrt{19}}$ (D) $f(3) = \frac{2}{\sqrt{19}} \tan^{-1} \frac{3}{\sqrt{19}}$

Ans. AC

Sol.

I02/2/057. Which of the following options is/are correct ?

(A) If $f(x) = \sqrt{x}$, and $g(x) = x^{\frac{1}{3}}$, then $\int_0^2 (f(1+x^3) + g(x^2+2x)) dx = 6$

(B) Let $f : (-1, 1) \rightarrow \mathbb{R}$ be a twice differentiable function such that $2f'(x) + xf''(x) = 1 \forall x$, then

$$\int_{-1}^1 x^3 f(x) dx = \frac{1}{5}, \text{ if } f'\left(\frac{1}{2}\right) = \frac{1}{2}$$

(C) If f is a function satisfying

$$f(x) = x + \int_0^1 (x+t)f(t) dt, \text{ then } \int_0^1 f(x) dx = -6$$

(D) If $f(x) = \ln x$ then

$$\int_1^{e^2} \frac{f(xe^{x+1})}{(x+1)^2 + f^2(x^x)} dx > \frac{\pi}{4}$$

निम्न में से कौनसा/से विकल्प सत्य होगा/होंगे ?

(A) यदि $f(x) = \sqrt{x}$, तथा $g(x) = x^{\frac{1}{3}}$, तब $\int_0^2 (f(1+x^3) + g(x^2+2x)) dx = 6$

(B) माना $f : (-1, 1) \rightarrow \mathbb{R}$ एक द्विअवकलनीय फलन इस प्रकार है कि $2f'(x) + xf''(x) = 1 \forall x$, तब

$$\int_{-1}^1 x^3 f(x) dx = \frac{1}{5}, \text{ if } f'\left(\frac{1}{2}\right) = \frac{1}{2}$$

(C) यदि फलन f , $f(x) = x + \int_0^1 (x+t)f(t) dt$ को संतुष्ट करता है, तब $\int_0^1 f(x) dx = -6$

(D) यदि $f(x) = \ln x$ तब

$$\int_1^{e^2} \frac{f(xe^{x+1})}{(x+1)^2 + f^2(x^x)} dx > \frac{\pi}{4}$$

Ans. ABD

Sol.

I03/2/058. Let $y = f(x)$ be a real valued function satisfying $x \frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$, then :

(A) $f(x)$ is minimum at $x = 1$

(B) $f(x)$ is maximum at $x = 1$

(C) $f(3) = 5$

(D) $f(2) = 3$

माना $y = f(x)$ एक वास्तविक मान फलन है जो $x \frac{dy}{dx} = x^2 + y - 2$, $f(1) = 1$ को संतुष्ट करता है, तब :

(A) $x = 1$ पर $f(x)$ न्यूनतम है

(B) $x = 1$ पर $f(x)$ अधिकतम है

(C) $f(3) = 5$

(D) $f(2) = 3$

Ans. AC

Sol. $\frac{xdy - ydx}{x^2} = \frac{x^2 - 2}{x^2} dx$

$$\int d\left(\frac{y}{x}\right) = \int \left(1 - \frac{2}{x^2}\right) dx$$

$$y = x^2 - 2x + 2 \quad (\because f(1) = 1)$$

$$\frac{dy}{dx} = 2x - 2 = 0 \Rightarrow x = 1$$

$$\frac{d^2y}{dx^2} > 0 \Rightarrow \text{at } x = 1, f(x) \text{ will be minimum.}$$

I03/2/059. Let $y = f(x)$ defined on $\mathbb{R}$ satisfies the differential equation $(1 + x^2) \frac{dy}{dx} + 2xy = 2x$ and $f(0) = 2$, then which

of the following statement(s) is(are) correct :

(A) $f(x)$ is neither even nor odd function

(B) $f(x)$ increases on $(-\infty, 0)$ and decreases on $(0, \infty)$

(C) The x-intercept of normal on the graph of $y = f(x)$ at $x = 1$ equals $\frac{1}{4}$

(D) The area bounded by $y = f(x)$ with x-axis between ordinates at $x = 0$ and $x = 1$ equals $\left(\frac{\pi + 4}{4}\right)$

यदि फलन $y = f(x)$, $\mathbb{R}$ में इस प्रकार परिभाषित है कि अवकल समीकरण $(1 + x^2) \frac{dy}{dx} + 2xy = 2x$ को संतुष्ट करता है तथा

$f(0) = 2$ है, तो निम्न में से कौनसा/कौनसे कथन सत्य है :

(A) $f(x)$ ना तो सम, ना ही विषम

(B) $f(x)$, $(-\infty, 0)$ में वर्धमान तथा $(0, \infty)$ में हासमान

(C) फलन $y = f(x)$ के ग्राफ पर $x = 1$ पर अभिलम्ब का x-अन्तःखण्ड $\frac{1}{4}$ है

(D) फलन $y = f(x)$, x-अक्ष, $x = 0$ तथा $x = 1$ द्वारा परिबद्ध क्षेत्रफल $\left(\frac{\pi + 4}{4}\right)$ है

Ans. BCD

Sol. We have, $\frac{dy}{dx} + \left(\frac{2x}{1+x^2}\right)y = \left(\frac{2x}{1+x^2}\right)$ (Linear differential equation)

So, I.F. = $e^{\int \frac{2x}{1+x^2} dx} = e^{\ln(1+x^2)} = (1+x^2)$

Now, general solution is

$$y(1+x^2) = \int \left(\frac{2x}{1+x^2} \right) \cdot (1+x^2) dx + C$$

$$\therefore y(1+x^2) = x^2 + C \quad \dots\dots(1)$$

As $y(0) = 2 \Rightarrow 2 = C$

$$\therefore f(x) = \left(\frac{x^2 + 2}{x^2 + 1} \right) = 1 + \frac{1}{x^2 + 1} \text{ (even function)}$$

Range of $f(x) = (1, 2]$

at $x = 1, y = \frac{3}{2}$

$$\frac{dy}{dx} = \frac{-2x}{(1+x^2)^2}$$

$$\text{slope of normal} = \frac{(1+x^2)^2}{2x}$$

Equation of normal

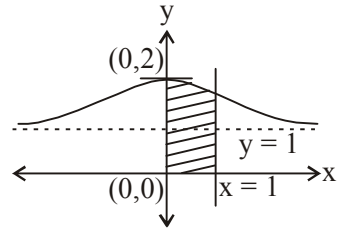
$$y - \frac{3}{2} = \frac{(1+1)^2}{2} (x-1)$$

$$y - \frac{3}{2} = 2(x-1)$$

For x-intercept $y = 0$

$$x = 1 - \frac{3}{4} = \frac{1}{4}$$

$$\text{Also, Area} = \int_0^1 \left(1 + \frac{1}{1+x^2} \right) dx = (x + \tan^{-1} x)_0^1 = 1 + \frac{\pi}{4} = \frac{\pi+4}{4}$$



I03/2/060. Let $f(x)$ be a differentiable function satisfying $\int_0^{f(x)} f^{-1}(t) dt - \int_0^x (\cos t - f(t)) dt = 0$ and $f(0) = 1$, then the

number of solution(s) of the equation $\left| \frac{f(2x)}{\sin x} - \frac{f(x)}{2} \right| = 0$ in $(0, 2\pi)$ are less than :

माना $f(x)$ अवकलनीय फलन है तथा $\int_0^{f(x)} f^{-1}(t)dt - \int_0^x (\cos t - f(t))dt = 0$ को संतुष्ट करता है तथा $f(0) = 1$, तो

अन्तराल $(0, 2\pi)$ में समीकरण $\left| \frac{f(2x)}{\sin x} - \frac{f(x)}{2} \right| = 0$ के हलों की संख्या निम्न में से किससे कम होगी :

(A) 2

(B) 3

(C) 4

(D) 5

Ans. CD

Sol. $\int_0^{f(x)} f^{-1}(t).dt - \int_0^x (\cos t - f(t)).dt = 0$

$$x.f(x) - \cos x + f(x) = 0$$

$$x \cdot \frac{dy}{dx} + y = \cos x$$

$$\int d(xy) = \int \cos x \cdot dx$$

$$\Rightarrow xy = \sin x \quad (\because f(0) = 1)$$

$$\left| \frac{\sin(2x)}{2x} - \frac{\sin x}{x} \right| = 0$$

$$\frac{\sin(2x)}{x} - \frac{\sin^2 x}{x} = 0$$

$$= \frac{\sin x}{x} \{2 \cos x - \sin x\} = 0$$

$$\frac{\sin x}{x} \neq 0 \Rightarrow \cot x = \frac{1}{2}$$

I03/2/061. Solution of the differential equation $(2 + 2x^2\sqrt{y})ydx + (x^2\sqrt{y} + 2)x dy = 0$ is/are :

अवकलनीय समीकरण $(2 + 2x^2\sqrt{y})ydx + (x^2\sqrt{y} + 2)x dy = 0$ का हल होगा/होंगे :

(A) $xy(x^2\sqrt{y} + 5) = c$ (B) $xy(x^2\sqrt{y} + 3) = c$ (C) $xy(y^2\sqrt{x} + 3) = c$ (D) $xy(y^2\sqrt{x} + 5) = c$

Ans. B

Sol. $2ydx + 2xdy + (2x^2y^{3/2}dx + x^3y^{1/2}dy) = 0$

$$2d(xy) + \frac{2}{3}d(x^3 \times y^{3/2}) = 0$$

On integrating

$$xy + \frac{1}{3} \left(x^3 \cdot y^{\frac{3}{2}} \right) = 0 \Rightarrow xy(3 + x^2 \sqrt{y}) = 0$$

I03/2/062. The solution of the differential equation $(y^2 dx - 2xy dy) = x^3 y^3 dy + x^2 y^4 dx$ is :

अवकलनीय समीकरण $(y^2 dx - 2xy dy) = x^3 y^3 dy + x^2 y^4 dx$ का हल है :

(A) $\log \left(\frac{x}{y^2} \right) = \frac{(xy)^2}{2} + c$

(B) $\frac{x}{cy^2} = e^{\frac{(xy)^2}{2}}$

(C) $\left(\frac{x}{y^2} \right) = 2 \log(xy) + c$

(D) $\left(\frac{x}{y^2} \right) = e^{2(xy)} + c$

Ans. AB

Sol. $\frac{y^2 dx - 2xy dy}{y^4} = \frac{x^2}{y} (x dy + y dx)$

$$\frac{y^2}{x} \left(\frac{y^2 dx - 2xy dy}{y^4} \right) = xy (x dy + y dx)$$

$$\int d \left(\log \left(\frac{y^2}{x} \right) \right) = \int xy d(xy)$$

$$\log \left(\frac{y^2}{x} \right) = \frac{(xy)^2}{2} + c$$

I03/2/063. Solution of the differential equation $(3 \tan x + 4 \cot y - 7) \sin^2 y dx - (4 \tan x + 7 \cot y - 5) \cos^2 x dy = 0$ is :

अवकलनीय समीकरण $(3 \tan x + 4 \cot y - 7) \sin^2 y dx - (4 \tan x + 7 \cot y - 5) \cos^2 x dy = 0$ का हल होगा :

(A) $\frac{3}{2} \cot^2 x - 7 \cot x + \frac{7}{2} \tan^2 y - 5 \tan y + 4 \cot x \cdot \tan y = c$

(B) $\frac{3}{2} \tan^2 x - 7 \tan x + \frac{7}{2} \cot^2 y - 5 \cot y + 4 \tan x \cdot \cot y = c$

(C) $3 \tan^2 y - 14 \cot x \cdot \tan^2 y + 7 \cot^2 x - 10 \tan y \cot^2 x + 8 \cot x \cdot \tan y + 2c \cot^2 x \tan^2 y = 0$

(D) $3 \cot^2 y - 14 \cot x \cdot \cot^2 y + 7 \cot^2 x + 10 \cot y \tan^2 x + 8 \tan x \cdot \cot y = 0$

Ans. BC

Sol. Dividing throughout by $\cos^2 x \sin^2 y$, the given differential equation becomes

$$(3 \tan x + 4 \cot y - 7) \sec^2 x dx - (4 \tan x + 7 \cot y - 5) \operatorname{cosec}^2 y dy = 0$$

$$\Rightarrow 3 \tan x \sec^2 x \, dx - 7 \sec^2 x \, dx - 7 \cot y \operatorname{cosec}^2 y \, dy + 5 \operatorname{cosec}^2 y \, dy + 4 \cot y \sec^2 x \, dx - 4 \tan x \operatorname{cosec}^2 y \, dy = 0$$

$$\Rightarrow 3 \tan x \, d(\tan x) - 7 \, d(\tan x) + 7 \cot y \, d(\cot y) - 5 \, d(\cot y) + 4 \, d(\tan x \cot y) = 0$$

On integrating, we obtain

$$\frac{3}{2} \tan^2 x - 7 \tan x + \frac{7}{2} \cot^2 y - 5 \cot y + 4 \tan x \cot y = C$$

I03/2/064. Solution of the differential equation : $\frac{x+y \frac{dy}{dx}}{y-x \frac{dy}{dx}} = \frac{x \sin^2(x^2+y^2)}{y^3}$:

अवकलनीय समीकरण $\frac{x+y \frac{dy}{dx}}{y-x \frac{dy}{dx}} = \frac{x \sin^2(x^2+y^2)}{y^3}$ का हल होगा :

(A) $-\cot(x^2+y^2) = \left(\frac{x}{y}\right)^2 + c$

(B) $\frac{y^2}{x^2+y^2} = -\tan(x^2+y^2)$

(C) $-\cot(x^2+y^2) = \left(\frac{y}{x}\right)^2 + c$

(D) $\frac{y^2+x^2c}{x^2} = -\tan(x^2+y^2)$

Ans. AB

Sol. The given differential equation can be written as

$$\frac{x dx + y dy}{y dx - x dy} = \frac{x \sin^2(x^2+y^2)}{y^3}$$

$$\Rightarrow \frac{2x dx + 2y dy}{y dx - x dy} = \frac{2x \sin^2(x^2+y^2)}{y^3} \Rightarrow \frac{d(x^2+y^2)}{\sin^2(x^2+y^2)} = \frac{2x}{y^3} (y dx - x dy)$$

$$\Rightarrow \operatorname{cosec}^2(x^2+y^2) d(x^2+y^2) = 2 \left(\frac{x}{y}\right) d\left(\frac{x}{y}\right)$$

On integrating, we get

$$\int \operatorname{cosec}^2(x^2+y^2) d(x^2+y^2) = 2 \int \left(\frac{x}{y}\right) d\left(\frac{x}{y}\right)$$

$$\Rightarrow -\cot(x^2 + y^2) = \left(\frac{x}{y}\right)^2 + C, \text{ which is the required solution.}$$

I01/2/065. If $f(x) = \int \frac{x}{2} d\left(\frac{x^2-1}{x^2}\right)$ and $f(2) = \frac{1}{2}$ and $g_r(x) = \underbrace{f[f\{f(\dots f(x)\dots)\}]}_{r\text{-times}}$ i.e. $g_1(x) = f(x)$, $g_2(x) = f[f(x)]$

and so on then identify the correct statement(s) :

(A) $\frac{d}{dx}(g_{(3n-2)}(x)) = 1$ whenever exists, $n \in \mathbb{N}$

(B) $\frac{d}{dx}(g_{(3n)}(x)) = 1$ whenever exists, $n \in \mathbb{N}$

(C) $\lim_{x \rightarrow 1} \frac{\sum_{r=1}^{100} (g_{3r}(x))^r + x - 101}{x - 1} = 5050$

(D) Slope of the tangent to the graph of the function $y = g_{80}(x)$ at $x = \frac{1}{2}$ is 4

यदि $f(x) = \int \frac{x}{2} d\left(\frac{x^2-1}{x^2}\right)$ तथा $f(2) = \frac{1}{2}$ तथा $g_r(x) = \underbrace{f[f\{f(\dots f(x)\dots)\}]}_{r\text{-times}}$ i.e. $g_1(x) = f(x)$, $g_2(x) = f[f(x)]$

तथा यह क्रम इसी प्रकार चलता रहेगा, तो निम्न में से कौनसे कथन सत्य होंगे :

(A) $\frac{d}{dx}(g_{(3n-2)}(x)) = 1$ जब भी विद्यमान हो, $n \in \mathbb{N}$

(B) $\frac{d}{dx}(g_{(3n)}(x)) = 1$ जब भी विद्यमान हो, $n \in \mathbb{N}$

(C) $\lim_{x \rightarrow 1} \frac{\sum_{r=1}^{100} (g_{3r}(x))^r + x - 101}{x - 1} = 5050$

(D) फलन $y = g_{80}(x)$, $x = \frac{1}{2}$ पर स्पर्श रेखा का ढाल 4 होगा

Ans. BD

Sol. $f(x) = \int \frac{x}{2} d\left(1 - \frac{1}{x^2}\right) = \int \frac{x}{2} \cdot \frac{2}{x^3} dx$

$$= \int \frac{1}{x^2} dx = -\frac{1}{x} + C$$

$$f(2) = -\frac{1}{2} + C = \frac{1}{2} \Rightarrow C = 1$$

$$\therefore f(x) = \frac{x-1}{x}$$

$$g_1(x) = f(x) = \frac{x-1}{x}; \quad g_2(x) = f(f(x)) = \frac{\frac{x-1}{x} - 1}{\frac{x-1}{x}} = \frac{-1}{x-1} = \frac{1}{1-x};$$

$$g_3(x) = f(f(f(x))) = \frac{1}{1 - \left(\frac{x-1}{x}\right)} = x$$

$$g_4(x) = g_1(x) = \frac{x-1}{x}; \quad g_5(x) = g_2(x) = \frac{1}{1-x}; \quad g_6(x) = g_3(x) = x \text{ and so on}$$

$$g_{(3n-2)}(x) = \frac{x-1}{x}, \quad g_{(3n-1)}(x) = \frac{1}{1-x}, \quad g_{3n}(x) = x$$

$$\frac{d}{dx}(g_{(3n-2)}(x)) = \frac{d}{dx}\left(1 - \frac{1}{x}\right) = \frac{1}{x^2}$$

Option (A) is wrong

$$\frac{d}{dx}(g_{(3n)}(x)) = \frac{d}{dx}(x) = 1$$

Option (B) is correct

$$\lim_{x \rightarrow 1} \frac{x + x^2 + \dots + x^{100} + x - 101}{x - 1}$$

Applying L'Hospital's rule

$$\lim_{x \rightarrow 1} \frac{1 + 2x + \dots + 100x^{99} + 1}{1} = 1 + 2 + \dots + 100 + 1 = 5051 \text{ Option (C) is wrong}$$

$$y = g_{80}(x) = g_{(3n-1)}(x) = \frac{1}{1-x}$$

$$\therefore g'_{(3n-1)}(x) = \frac{1}{(1-x)^2}$$

$$g'_{(3n-1)}(x) \Big|_{x=\frac{1}{2}} = 4$$

Option (D) is correct

101/2/066. If $f(x) = \lim_{n \rightarrow \infty} e^{x \tan\left(\frac{1}{n}\right) \ln\left(\frac{1}{n}\right)}$ and $\int \frac{f(x)}{\sqrt[3]{(\sin^{-1} x \cos x)}} dx = g(x) + c$, $g\left(-\frac{\pi}{4}\right) = -\frac{15}{8}$, then :

$$(A) \quad g\left(\frac{\pi}{4}\right) = \frac{3}{2}$$

(B) $g(x)$ is continuous for all x

$$(C) \quad g\left(\frac{\pi}{4}\right) = -\frac{15}{8}$$

(D) $g(x)$ is non differentiable at infinitely many points

यदि $f(x) = \lim_{n \rightarrow \infty} e^{x \tan\left(\frac{1}{n}\right) \ln\left(\frac{1}{n}\right)}$ तथा $\int \frac{f(x)}{\sqrt[3]{(\sin^{11} x \cos x)}} dx = g(x) + c$, $g\left(-\frac{\pi}{4}\right) = -\frac{15}{8}$, तब :

$$(A) \quad g\left(\frac{\pi}{4}\right) = \frac{3}{2}$$

(B) $g(x)$, सभी x के लिए सतत् है

$$(C) \quad g\left(\frac{\pi}{4}\right) = -\frac{15}{8}$$

(D) $g(x)$ अनन्त बिन्दुओं पर अवकलनीय नहीं है

Ans.

CD

Sol.

$$\therefore \lim_{n \rightarrow \infty} \tan\left(\frac{1}{n}\right) \ln\left(\frac{1}{n}\right) = \lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)} \cdot \frac{\ln\left(\frac{1}{n}\right)}{1} = \lim_{n \rightarrow \infty} \frac{\tan\left(\frac{1}{n}\right)}{\left(\frac{1}{n}\right)} \cdot \frac{\ln(n)}{n} = -1 \cdot \lim_{n \rightarrow \infty} \frac{\left(\frac{1}{n}\right)}{1} = 0$$

Then, $f(x) = e^0 = 1$

$$\therefore \int \frac{f(x)}{\sqrt[3]{(\sin^{11} x \cos x)}} dx = \int \frac{1}{\sin^{\frac{11}{3}} x \cos^{\frac{1}{3}} x} dx = \int \sin^{-\frac{11}{3}} x \cdot \cos^{-\frac{1}{3}} x dx$$

$$= \int (\tan x)^{-\frac{11}{3}} \cos^{-4} x dx = \int (\tan x)^{-\frac{11}{3}} \cdot \sec^4 x dx = \int (\tan x)^{-\frac{11}{3}} \cdot (1 + \tan^2 x) \cdot \sec^2 x dx$$

$$= \frac{(\tan x)^{-\frac{11}{3}+1}}{\left(-\frac{11}{3}+1\right)} + \frac{(\tan x)^{-\frac{2}{3}}}{\left(-\frac{2}{3}\right)} + c = -\frac{3}{8}(\tan x)^{-\frac{8}{3}} - \frac{3}{2}(\tan x)^{-\frac{2}{3}} + c$$

$$\therefore g(x) = -\frac{3}{8}(\tan x)^{-\frac{8}{3}} - \frac{3}{2}(\tan x)^{-\frac{2}{3}}$$

$$\therefore g\left(\frac{\pi}{4}\right) = -\frac{3}{8} - \frac{3}{2} = -\frac{15}{8} \text{ and } g(x) \text{ is non differentiable at } \tan x = 0 \text{ or } x = n\pi, n \in \mathbb{I}.$$

COMPREHENSION TYPE QUESTIONS

Paragraph for Question I04/3/067 to I04/3/069

Let $y'(x)\phi(x) - y(x)\phi'(x) + y^2(x) = 0$, $y(1) = 1$, $x \in \mathbb{R}$, $\phi(x)$ and $y(x)$ are non constant differentiable function on $\mathbb{R}$ with $\phi(1) = 0$ and $\phi(2) = 4$.

माना $y'(x)\phi(x) - y(x)\phi'(x) + y^2(x) = 0$, $y(1) = 1$, $x \in \mathbb{R}$, $\phi(x)$ तथा $y(x)$, $\mathbb{R}$ पर अनियत अवकलनीय फलन हैं, $\phi(1) = 0$ तथा $\phi(2) = 4$.

I04/3/067. The value of $y(2)$ is equal to :

$y(2)$ का मान बराबर है :

- (A) 0 (B) 1 (C) 2 (D) 4

Ans. D

Sol. $\therefore y'(x)\phi(x) - y(x)\phi'(x) + y^2(x) = 0$

$$\Rightarrow \frac{d}{dx} \left(\frac{\phi(x)}{y(x)} \right) = 1 \Rightarrow \frac{\phi(x)}{y(x)} = x - 1 \quad \{ \because \phi(1) = 0 \}$$

$$\therefore \text{for } x = 2 \Rightarrow y(2) = \phi(2) = 4$$

I04/3/068. The value of $\int_1^2 \left(\frac{\phi(x)}{y(x)} \right) \frac{1}{x^2 \sqrt{x^2 + \left(\frac{\phi(x)}{y(x)} \right)^2}} dx$ is equal to :

$$\int_1^2 \left(\frac{\phi(x)}{y(x)} \right) \frac{1}{x^2 \sqrt{x^2 + \left(\frac{\phi(x)}{y(x)} \right)^2}} dx \text{ का मान होगा :}$$

- (A) $\sqrt{5}$ (B) $\frac{\sqrt{5}}{2} - 1$ (C) $\sqrt{5} - 1$ (D) $\sqrt{5} - 2$

Ans. B

Sol. $\int_1^2 \frac{x-1}{x^2 \sqrt{x^2 + (x-1)^2}} dx = \int_1^2 \frac{x \left(x - \frac{1}{x} \right)}{x^2 \cdot x \sqrt{1 + \left(1 - \frac{1}{x} \right)^2}} = \int_0^{\frac{1}{2}} \frac{t}{\sqrt{1+t^2}} dt$ where $t = 1 - \frac{1}{x} = \left[\sqrt{1+t^2} \right]_0^{\frac{1}{2}} = \frac{\sqrt{5}}{2} - 1$

I04/3/069. Which one of the following statement is **correct** for function $g(x) = \frac{\phi(x)}{y(x)}$:

- (A) $g(x)$ is decreasing on R (B) Area bounded by $g(x)$ and coordinate axes is $\frac{1}{4}$
 (C) $g(x)$ has minima but no maxima (D) $|g(x)|$ is continuous but not differentiable on R

फलन $g(x) = \frac{\phi(x)}{y(x)}$ के लिए निम्न कथनों में से कौनसा कथन सत्य होगा :

- (A) $g(x)$, R पर ह्रासमान है (B) $g(x)$ तथा निर्देशांक अक्षों से परिबद्ध क्षेत्रफल $\frac{1}{4}$ है
 (C) $g(x)$ का निम्निष्ठ है लेकिन उच्चिष्ठ नहीं (D) $|g(x)|$, R पर सतत् है लेकिन अवकलनीय नहीं

Ans. D

Sol. $g(x) = \frac{\phi(x)}{y(x)} = x - 1$

Paragraph for Question I04/3/070 to I04/3/071

The tangent at point P of a curve $y = f(x)$ meets the y-axis at A, x-axis at B and the line parallel to y-axis through P meets x-axis at C such that area of ΔPBC is equal to the ratio of ordinate to abscissae of point P where $f(1) = 1$ and $f(3) = e^2$.

वक्र $y = f(x)$ के बिन्दु P पर स्पर्शरेखा y-अक्ष को A, x-अक्ष को B तथा P से y-अक्ष के समानान्तर गुजरने वाली रेखा x-अक्ष के बिन्दु C पर इस प्रकार मिलती है कि ΔPBC का क्षेत्रफल बिन्दु P की कोटि तथा भुज के अनुपात के बराबर है जहाँ $f(1) = 1$ तथा $f(3) = e^2$ है।

I04/3/070. If the area bounded by the curve $y = f(x)$ line $x + y = 2$ and y-axis in the Ist quadrant is A, then :

प्रथम चतुर्थांश में वक्र $y = f(x)$, रेखा $x + y = 2$ तथा y-अक्ष से परिबद्ध क्षेत्रफल A है, तब :

- (A) $\frac{1}{2} < A < 1$ (B) $1 < A < 2$ (C) $2 < A < 3$ (D) $3 < A < 4$

Ans. A

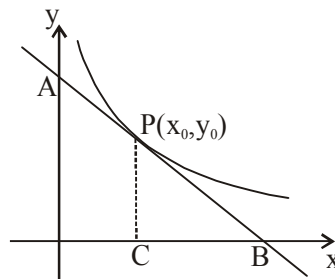
Sol.

I04/3/071. The value of $\int_{-2013}^{2013} \text{sgn}(f(x - 2013)) dx$ is (where $\text{sgn}(\cdot)$ denotes signum function) :

$\int_{-2013}^{2013} \text{sgn}(f(x - 2013)) dx$ का मान है (जहाँ $\text{sgn}(\cdot)$ सिग्नम फलन को दर्शाता है) :

- (A) 0 (B) 2013 (C) 2026 (D) 4026

Ans. D



Sol. Equ. of tangent at $P(x_0, y_0)$

$$y - y_0 = f'(x_0)(x - x_0)$$

$$B\left(x_0 - \frac{y_0}{f'(x_0)}, 0\right)$$

$$\text{if } \text{Ar}(\triangle PBC) = \frac{y_0}{x_0}$$

$$\left| \frac{1}{2} x \left(\frac{-y_0}{f'(x_0)} \right) \cdot y_0 \right| = \frac{y_0}{x_0} \Rightarrow \frac{dy}{dx} = \frac{xy}{2} \Rightarrow y = e^{\frac{x^2}{4} - \frac{1}{4}} \quad (\because f(1) = 1 \text{ and } f(3) = e^2)$$

Sol-1.

$$A = \int_0^1 \left((2-x) - e^{\frac{x^2}{4} - \frac{1}{4}} \right) dx = \frac{3}{2} - \int_0^1 e^{\frac{x^2}{4} - \frac{1}{4}} \cdot dx ;$$

$$e^{-\frac{1}{4}} < \int_0^1 e^{\frac{x^2}{4} - \frac{1}{4}} \cdot dx < 1 \Rightarrow \frac{1}{2} < A < 1$$

Sol-2.

$$\int_{-2013}^{2013} dx = 4026$$

Paragraph for Question I04/3/072 to I04/3/074

If the curve $y = f(x)$ satisfy the equation $y(x + y^3)dx = x(y^3 - x)dy$ and

$$g(x) = \int_{\frac{1}{8}}^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_{\frac{1}{8}}^{\cos^2 x} \cos^{-1} \sqrt{t} dt, \text{ where } x \in \left[0, \frac{\pi}{2}\right], \text{ then :}$$

यदि वक्र $y = f(x)$ अवकलज समीकरण $y(x + y^3)dx = x(y^3 - x)dy$ को संतुष्ट करता है तथा

$$g(x) = \int_{\frac{1}{8}}^{\sin^2 x} \sin^{-1} \sqrt{t} dt + \int_{\frac{1}{8}}^{\cos^2 x} \cos^{-1} \sqrt{t} dt, \text{ जहाँ } x \in \left[0, \frac{\pi}{2}\right], \text{ तब :}$$

I04/3/072. Equation of curve $y = f(x)$ passing through $(4, -2)$ is :

$(4, -2)$ से गुजरने वाले वक्र $y = f(x)$ का समीकरण होगा :

$$(A) 3y = (-54x)^{\frac{1}{3}} \quad (B) 2y = (-16x)^{\frac{1}{3}} \quad (C) y = (-2x)^{\frac{1}{3}} \quad (D) 4y = (-12x)^{\frac{1}{3}}$$

Ans. C

Sol. $y(x + y^3)dx = x(y^3 - x)dy$

$$\Rightarrow \frac{d(xy)}{(xy)^2} = \left(\frac{y}{x}\right) d\left(\frac{y}{x}\right) \Rightarrow y = (-2x)^{\frac{1}{3}}$$

I04/3/073. Area bounded by curve $y = f(x)$, $g(x)$ and y-axis, is :

वक्र $y = f(x)$, $g(x)$ तथा y-से परिबद्ध क्षेत्रफल है :

(A) $\frac{1}{4}\left(\frac{3\pi}{16}\right)^4$ (B) $\frac{1}{8}\left(\frac{3\pi}{16}\right)^4$ (C) $\frac{1}{4}\left(\frac{3\pi}{8}\right)^4$ (D) $\frac{1}{8}\left(\frac{3\pi}{8}\right)^4$

Ans. B

Sol. Differentiating $g(x)$, we get $g'(x) = 0$
 $\Rightarrow g(x)$ is a constant function.

also putting $\sin x = \cos x = \frac{1}{\sqrt{2}}$ we get $g(x) = \frac{3\pi}{16}$

I04/3/074. The solution of the equation $f(x) = g(x)$ is :

समीकरण $f(x) = g(x)$ का हल होगा :

(A) $-\frac{1}{2}\left(\frac{3\pi}{16}\right)^3$ (B) $\frac{1}{2}\left(\frac{3\pi}{16}\right)^3$ (C) $-\frac{1}{2}\left(\frac{3\pi}{8}\right)^3$ (D) $\frac{1}{2}\left(\frac{3\pi}{8}\right)^3$

Ans. A

Sol. $(-2x)^{\frac{1}{3}} = \frac{3\pi}{16}$

$$\Rightarrow -2x = \left(\frac{3\pi}{16}\right)^3 \Rightarrow x = -\frac{1}{2}\left(\frac{3\pi}{16}\right)^3$$

Paragraph for Question I02/3/075 to I02/3/077

Let $f(x)$ be function defined on $[0, 1]$ such that $f(1) = 0$ and for any $a \in (0, 1]$,

$$\int_0^a f(x) dx - \int_a^1 f(x) dx = 2f(a) + 3a + b \text{ where } b \text{ is constant.}$$

माना $f(x)$ एक फलन $[0, 1]$ में इस प्रकार परिभाषित है कि $f(1) = 0$ तथा किसी $a \in (0, 1]$ के लिए,

$$\int_0^a f(x) dx - \int_a^1 f(x) dx = 2f(a) + 3a + b \text{ जहाँ } b \text{ अचर है।}$$

I02/3/075. $b =$

(A) $\frac{3}{2e} - 3$

(B) $\frac{3}{2e} - \frac{3}{2}$

(C) $\frac{3}{2e} + 3$

(D) $\frac{3}{2e} + \frac{3}{2}$

Ans. A

Sol. $\int_0^a f(x) dx - \int_a^1 f(x) dx = 2f(a) + 3a + b \quad \dots(1)$

diff. w.r.t. 'a' on both sides

$$f(a) - 0 - (0 - f(a)) = 2f'(a) + 3$$

$$2f(a) = 2f'(a) + 3$$

$$(2f(a) - 3) = 2f'(a)$$

$$\frac{2f'(a)}{2f(a) - 3} = 1$$

$$\int \frac{2f'(a)}{2f(a) - 3} da = \int da$$

$$\ln |2f(a) - 3| = a + c$$

$$2f(a) - 3 = e^{a+c}$$

$$2f(a) - 3 = ke^a$$

$$2f(a) = ke^a + 3$$

$$\text{put } a = 1 \Rightarrow 0 = ke + 3 \Rightarrow k = -\frac{3}{e}$$

$$\therefore 2f(a) = -\frac{3}{e} e^a + 3$$

$$f(a) = \frac{3}{2} - \frac{3}{2e} e^a$$

$$f(x) = \frac{3}{2} - \frac{3}{2e} e^x$$

put $f(x)$ in (1) (By taking limiting case)

$$\int_0^a \left(\frac{3}{2} - \frac{3}{2e} e^x \right) dx - \int_a^1 \left(\frac{3}{2} - \frac{3}{2e} e^x \right) dx = 3 - \frac{3}{e} e^a + 3a + b$$

$$\left[\left(\frac{3a}{2} - \frac{3}{2e} e^a \right) - \left(0 - \frac{3}{2e} \right) \right] - \left[\left(\frac{3}{2} - \frac{3}{2} \right) - \left(\frac{3a}{2} - \frac{3e^a}{2e} \right) \right] = 3 - \frac{3}{e} e^a + 3a + b$$

$$\frac{3}{2e} - 3 = b$$

I02/3/076. The length of the subtangent of the curve $y = f(x)$ at $x = 1/2$ is :

चक्र $y = f(x)$ के बिन्दु $x = 1/2$ पर अधोसपरेन्ट की लम्बाई होगी :

- (A) $\sqrt{e}-1$ (B) $\frac{\sqrt{e}-1}{2}$ (C) $\sqrt{e}+1$ (D) $\frac{\sqrt{e}+1}{2}$

Ans. A

Sol. Length of subtangent = $\left| \frac{y}{\left(\frac{dy}{dx}\right)} \right|$

$$y = f(x) = \frac{3}{2} - \frac{3}{2e} e^x$$

$$\frac{dy}{dx} = f'(x) = 0 - \frac{3}{2e} e^x$$

$$\left. \frac{dy}{dx} \right|_{x=1/2} = -\frac{3}{2\sqrt{e}}$$

$$\text{when } x = 1/2 \quad y = f(1/2) = \frac{3}{2} - \frac{3}{2e} e^{1/2} = \frac{3}{2} \left(1 - \frac{1}{\sqrt{e}} \right)$$

$$\text{Length of subtangent} = \left| \frac{\frac{3}{2} \left(1 - \frac{1}{\sqrt{e}} \right)}{-\frac{3}{2\sqrt{e}}} \right| = \left| \sqrt{e} - 1 \right| = \sqrt{e} - 1$$

I02/3/077. $\int_0^1 f(x) dx = :$

- (A) $\frac{1}{e}$ (B) $\frac{1}{2e}$ (C) $\frac{3}{2e}$ (D) $\frac{2}{e}$

Ans. C

Sol. $\int_0^1 f(x) dx = \int_0^1 \left(\frac{3}{2} - \frac{3}{2e} e^x \right) dx = \left. \frac{3x}{2} - \frac{3}{2e} e^x \right|_0^1 = \left(\frac{3}{2} - \frac{3e}{2e} \right) - \left(0 - \frac{3}{2e} \right) = \left(\frac{3}{2} - \frac{3}{2} \right) + \frac{3}{2e} = \frac{3}{2e}$

Paragraph for Question I02/3/078 to I02/3/079

Given $f(x) = ax^2 + bx + c$ is a quadratic polynomial, where $a, b, c \in \mathbb{R}$ and $a + b = 0$, also

$$\int_0^1 (f(x) - \pi\sqrt{2} \sin \pi x) dx = 0 \text{ and } g(x) = \sqrt{x} \text{ then}$$

On the basis of above information, answer the following questions :

दिया गया है $f(x) = ax^2 + bx + c$ एक द्विघात बहुपद है जहाँ $a, b, c \in \mathbb{R}$ तथा $a + b = 0$ एवं

$$\int_0^1 (f(x) - \pi\sqrt{2} \sin \pi x) dx = 0 \text{ तथा } g(x) = \sqrt{x} \text{ हो तो}$$

उपरोक्त जानकारी के आधार पर निम्न प्रश्नों के उत्तर दीजिये :

I02/3/078. $\int_{\frac{1}{6}}^{5/6} \cos \pi x \cdot \sin(f(x)) \cdot \ln(1 + e^{\csc \pi x}) dx$ is not equal to :

(A) $\frac{1}{2} \int_{1/6}^{5/6} \cos \pi x \sin(f(x)) dx$

(B) $\int_{1/6}^{5/6} \cos \pi x \sin(f(x)) dx$

(C) $\int_{1/6}^{5/6} \cot \pi x \sin(f(x)) dx$

(D) None of these

$\int_{\frac{1}{6}}^{5/6} \cos \pi x \cdot \sin(f(x)) \cdot \ln(1 + e^{\csc \pi x}) dx$ is not equal to :

(A) $\frac{1}{2} \int_{1/6}^{5/6} \cos \pi x \sin(f(x)) dx$

(B) $\int_{1/6}^{5/6} \cos \pi x \sin(f(x)) dx$

(C) $\int_{1/6}^{5/6} \cot \pi x \sin(f(x)) dx$

(D) इनमें से कोई नहीं

Ans. D

Sol. x coordinate of vertex of $f(x) = -\frac{b}{2a} = \frac{1}{2}$.

$$\therefore f\left(\frac{1}{2} - x\right) = f\left(\frac{1}{2} + x\right) \Rightarrow f(1 - x) = f(x)$$

$$\text{Let } I = \int_{1/6}^{5/6} \cos \pi x \cdot \sin(f(x)) \cdot \ln(1 + e^{\csc \pi x}) dx$$

using king $x \rightarrow 1 - x$

$$I = \int_{1/6}^{5/6} \cos \pi x \cdot \sin(f(x)) \cdot \ln(1 + e^{\csc \pi x}) dx$$

$$\therefore I = -I \Rightarrow I = 0$$

also in options A, B, C each integral is zero by king property.

I02/3/079. $\int_0^1 \left(\frac{g(x) + g(1-x)}{1 + \sqrt{2}g(x)} \right) f(x) dx$ is equal to :

$$\int_0^1 \left(\frac{g(x) + g(1-x)}{1 + \sqrt{2}g(x)} \right) f(x) dx \text{ बराबर होगा :}$$

- (A) 0 (B) $\frac{1}{\sqrt{2}}$ (C) 2 (D) $\sqrt{2}$

Ans. C

Sol. Let $S = \int_0^1 \frac{\sqrt{x} + \sqrt{1-x}}{1 + \sqrt{2}x} f(x) dx \dots(1)$

using king $S = \int_0^1 \frac{\sqrt{x} + \sqrt{1-x}}{1 + \sqrt{2}(1-x)} f(x) dx \dots(2)$

$$(1) + (2)$$

$$\Rightarrow 2S = \int_0^1 (\sqrt{x} + \sqrt{1-x}) \left(\frac{1}{1 + \sqrt{2}x} + \frac{1}{\sqrt{2}(1-x)} \right) f(x) dx$$

$$\Rightarrow 2S = \sqrt{2} \int_0^1 f(x) dx = 2\pi \int_0^1 \sin \pi x dx = 4$$

$$\Rightarrow S = 2$$

Paragraph for Question I02/3/080 to I02/3/081

$f(x)$ satisfies the relation $f(x) - \lambda \int_0^{\pi/2} \sin x \cos t f(t) dt = \sin x$.

$f(x)$ संबंध $f(x) - \lambda \int_0^{\pi/2} \sin x \cos t f(t) dt = \sin x$ को संतुष्ट करता है।

I02/3/080. If $f(x) = 2$ has at least one real root, then :

यदि $f(x) = 2$ का कम से कम एक वास्तविक मूल है, तो :

- (A) $\lambda \in [1, 4]$ (B) $\lambda \in [-1, 2]$ (C) $\lambda \in [0, 1]$ (D) $\lambda \in [1, 3]$

Ans. D

Sol.

I02/3/081. If $\int_0^{\pi/2} f(x) dx = 3$, then the value of λ is :

- (A) 1 (B) $\frac{3}{2}$ (C) $\frac{4}{3}$ (D) none of these

यदि $\int_0^{\pi/2} f(x) dx = 3$ है, तो λ का मान होगा :

- (A) 1 (B) $\frac{3}{2}$ (C) $\frac{4}{3}$ (D) इनमें से कोई नहीं

Ans. B

Sol. $f(x) - \lambda \sin x \int_0^{\pi/2} \cos t f(t) dt = \sin x$

$$\Rightarrow f(x) - P \sin x = \sin x$$

$$\text{or } f(x) = (P + 1) \sin x$$

$$\text{Where } P = \lambda \int_0^{\pi/2} \cos t f(t) dt$$

$$= \lambda \int_0^{\pi/2} \cos t (P + 1) \sin t dt$$

$$= \frac{\lambda(P + 1)}{2} \int_0^{\pi/2} \sin 2t dt$$

$$P = \frac{\lambda(P + 1)}{2} \Rightarrow P = \frac{\lambda}{2 - \lambda}$$

$$f(x) = \left(\frac{\lambda}{2 - \lambda} + 1 \right) \sin x = 2$$

$$\Rightarrow \sin x = x = 2 - \lambda$$

$$\Rightarrow \lambda \in [1, 3]$$

$$\int_0^{\pi/2} \frac{2}{2 - \lambda} \sin x dx = 3$$

$$\Rightarrow \lambda = \frac{4}{3}$$

Paragraph for Question I02/3/082 to I02/3/084

Let $f(x)$ be function defined on $[0, 1]$ such that $f(1) = 0$ and for any $a \in (0, 1]$,

$$\int_0^a f(x) dx - \int_a^1 f(x) dx = 2f(a) + 3a + b \text{ where } b \text{ is constant.}$$

I02/3/082. $b = :$

(A) $\frac{3}{2e} - 3$

(B) $\frac{3}{2e} - \frac{3}{2}$

(C) $\frac{3}{2e} + 3$

(D) $\frac{3}{2e} + \frac{3}{2}$

Ans. A

Sol. $\int_0^a f(x) dx - \int_a^1 f(x) dx = 2f(a) + 3a + b \quad \dots(1)$

diff. w.r.t. 'a' on both sides

$$f(a) - 0 - (0 - f(a)) = 2f'(a) + 3$$

$$2f(a) = 2f'(a) + 3$$

$$(2f(a) - 3) = 2f'(a)$$

$$\frac{2f'(a)}{2f(a) - 3} = 1$$

$$\int \frac{2f'(a)}{2f(a) - 3} da = \int da$$

$$\ln |2f(a) - 3| = a + c$$

$$2f(a) - 3 = e^{a+c}$$

$$2f(a) - 3 = ke^a$$

$$2f(a) = ke^a + 3$$

$$\text{put } a = 1 \Rightarrow 0 = ke + 3 \Rightarrow k = -\frac{3}{e}$$

$$\therefore 2f(a) = -\frac{3}{e}e^a + 3$$

$$f(a) = \frac{3}{2} - \frac{3}{2e}e^a$$

$$f(x) = \frac{3}{2} - \frac{3}{2e}e^x$$

put $f(x)$ in (1) (By taking limiting case)

$$\int_0^a \left(\frac{3}{2} - \frac{3}{2e}e^x \right) dx - \int_a^1 \left(\frac{3}{2} - \frac{3}{2e}e^x \right) dx = 3 - \frac{3}{e}e^a + 3a + b$$

$$\left[\left(\frac{3a}{2} - \frac{3}{2e}e^a \right) - \left(0 - \frac{3}{2e} \right) \right] - \left[\left(\frac{3}{2} - \frac{3}{2} \right) - \left(\frac{3a}{2} - \frac{3e^a}{2e} \right) \right] = 3 - \frac{3}{e}e^a + 3a + b$$

$$\frac{3}{2e} - 3 = b$$

I02/3/083. The length of the subtangent of the curve $y = f(x)$ at $x = 1/2$ is :

- (A) $\sqrt{e} - 1$ (B) $\frac{\sqrt{e} - 1}{2}$ (C) $\sqrt{e} + 1$ (D) $\frac{\sqrt{e} + 1}{2}$

Ans. A

Sol. Length of subtangent = $\left| \frac{y}{\left(\frac{dy}{dx}\right)} \right|$

$$y = f(x) = \frac{3}{2} - \frac{3}{2e} e^x$$

$$\frac{dy}{dx} = f'(x) = 0 - \frac{3}{2e} e^x$$

$$\left. \frac{dy}{dx} \right|_{x=1/2} = -\frac{3}{2\sqrt{e}}$$

$$\text{when } x = 1/2 \quad y = f(1/2) = \frac{3}{2} - \frac{3}{2e} e^{1/2} = \frac{3}{2} \left(1 - \frac{1}{\sqrt{e}} \right)$$

$$\text{Length of subtangent} = \left| \frac{\frac{3}{2} \left(1 - \frac{1}{\sqrt{e}} \right)}{-\frac{3}{2\sqrt{e}}} \right| = \left| \sqrt{e} - 1 \right| = \sqrt{e} - 1$$

I02/3/084. $\int_0^1 f(x) dx = :$

- (A) $\frac{1}{e}$ (B) $\frac{1}{2e}$ (C) $\frac{3}{2e}$ (D) $\frac{2}{e}$

Ans. C

Sol. $\int_0^1 f(x) dx = \int_0^1 \left(\frac{3}{2} - \frac{3}{2e} e^x \right) dx = \left. \frac{3x}{2} - \frac{3}{2e} e^x \right|_0^1 = \left(\frac{3}{2} - \frac{3e}{2e} \right) - \left(0 - \frac{3}{2e} \right) = \left(\frac{3}{2} - \frac{3}{2} \right) + \frac{3}{2e} = \frac{3}{2e}$

Paragraph for Question I02/3/085 to I02/3/086

Let $f(x) = 1 + \int_0^1 (xe^y + ye^x) f(y) dy$ where x and y are independent variables.

माना $f(x) = 1 + \int_0^1 (xe^y + ye^x) f(y) dy$ जहाँ x तथा y स्वतंत्र चर हैं।

I02/3/085. If complete solution set of 'x' for which function $h(x) = f(x) + 3x$ is strictly increasing is $(-\infty, k)$ then $\left\lceil \frac{4}{3}e^k \right\rceil$

equals to : (where $[.]$ denotes greatest integer function) :

- (A) 1 (B) 2 (C) 3 (D) 4

Ans. C

Sol. $f(x) = 1 + x \int_0^1 e^y f(y) dy + e^x \int_0^1 y f(y) dy$

Let $A = \int_0^1 e^y f(y) dy, B = \int_0^1 y f(y) dy \Rightarrow f(x) = 1 + Ax + Be^x \Rightarrow A = \int_0^1 e^y (1 + Ay + Be^y) dy$

$$B = -\frac{2}{e+1}, A = -\frac{3}{2}$$

$$f'(x) + 3 > 0$$

$$\Rightarrow -\frac{3}{2} - \frac{2e^x}{(e+1)} + 3 > 0 \Rightarrow e^x < \frac{3(e+1)}{4} \Rightarrow \left\lceil \frac{4}{3}e^x \right\rceil = [e+1] = 3$$

I02/3/086. If acute angle of intersection of the curves $\frac{x}{2} + \frac{y}{3} + \frac{1}{3} = 0$ and $y = f(x)$ be θ then $\tan\theta$ equals to :

यदि वक्रों $\frac{x}{2} + \frac{y}{3} + \frac{1}{3} = 0$ तथा $y = f(x)$ का प्रतिच्छेदन न्यून कोण θ है तब $\tan\theta$ बराबर है :

- (A) $\frac{8}{25}$ (B) $\frac{16}{25}$ (C) $\frac{14}{25}$ (D) $\frac{4}{5}$

Ans. A

Sol. $Ax_1 + Be^{x_1} = -\frac{3x_1}{2} - 1 \Rightarrow Be^{x_1} = -1$

$$f'(x_1) = A + Be^{x_1} = m_1$$

$$m_1 = -\frac{3}{2} - 1 = -\frac{5}{2}$$